STUDY

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The EU - Japan Economic Partnership Agreement

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ABSTRACT

This report independently assesses the EU-Japan Economic Partnership Agreement. We find that the EPA establishes an ambitious framework to further liberalise and better organise trade, covering goods, services, intellectual property and investment, tariff- and non-tariff measures, and regulatory cooperation. Given its depth and breadth, and that it is unprecedented in including provisions on corporate governance, SMEs, and climate change, the EPA is set to become a benchmark for future trade agreements. Joining two open economies with high income levels and regulatory standards, the agreement is expected to generate benefits by boosting trade within sectors, minimising sectoral relocation and negative employment effects. Agri-food, textiles and leather products are where the EU can expect to make the greatest gains. Furthermore, the EPA will boost the EU's economic presence and political relevance in the Asia-Pacific area. Going beyond its economic benefits, the agreement also has significant non-economic implications. Reinforced cooperation will enhance the ability of both parties to shape the course of global developments in a manner that better reflects their shared interests and values, such as their commitment to a rule-based global trade system and the fight against global warming.

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List of Abbreviations

AI	Artificial Intelligence
CETA	EU-Canada Comprehensive Economic and Trade Agreement
CPTPP-11	Comprehensive and Progressive Agreement for Trans-Pacific Partnership
DTRI	Digital Trade Restrictiveness Index
EIF	Entry into Force
EUJEPA	EU-Japan Economic Partnership Agreement
EUKFTA	EU-South Korea Free Trade Agreement
EUSFTA	EU-Singapore Free Trade Agreement
GATS	General Agreement on Trade in Services
GDPR	General Data Protection Regulation
ICH	International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use
ICS	Investment Court System
ICT	Information and Communication Technology
ILO	International Labour Organisation
IoT	Internet of Things
ISDS	Investor-State Dispute Settlement
ISO	International Organization for Standardization
MFN	Most Favoured Nation
NTB	Non-Tariff Barriers
RCA	Revealed Comparative Advantage
SME	Small and Medium Enterprises
SPS	Sanitary and Phytosanitary measures
TBT	Technical Barriers to Trade
TSD	Trade and Sustainable Development
TTIP	Transatlantic Trade and Investment Partnership
UNECE	United Nations Economic Commission for Europe
UNFCCC	United Nations Framework Convention on Climate Change
WTO	World Trade Organisation

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1 Introduction

The EU-Japan Economic Partnership Agreement (EUJEPA) is the largest bilateral trade deal ever concluded by the EU in terms of market size, covering close to 30 % of global GDP. It includes commitments not only on trade in goods but also services and the promotion of bilateral investment. Moreover, it contains the most up-to-date EU provisions on corporate governance, the mobility of natural persons for business and Intellectual Property Rights (IPR) protection, including trade secrets. As such, it represents the joint efforts of the EU and Japan to develop models for new generation trade agreements.

According to the latest DG Trade estimates (European Commission, 2018), the impact of the agreement is expected to be balanced, with considerable gains for the EU in sectors such as agri-food, textiles and leather products. No EU sector would experience a significant loss. Overall, EUJEPA is expected to increase the EU's GDP by 0.14 % and total exports to Japan by EUR 13 billion by 2035. The main purpose of our study is to evaluate these main conclusions of DG Trade.

There are several reasons for the trade agreement between the EU and Japan. The decline of Japan's share of the EU's goods exports (from 6.9 % 1990 to 3.2 % in 2017) and likewise the share of Japanese goods in the EU's import basket (from 12 % in 1990 to 3.7 % in 2017) was a source of bilateral concern[1]. Currently, Japan is also not a major destination for European investment abroad, representing only 1.1 % of total extra-EU outward stock in 2016. The EUJEPA was therefore considered a strategic priority to revitalise trade and investment with Japan and expand opportunities for EU producers in the fast-growing Asian continent under the EU's Trade for All strategy launched in 2015.

The EU-Japan agreement has gained heightened geopolitical importance following the suspension in 2017 of talks between the EU and United States on the Transatlantic Trade and Investment Partnership (TTIP), and the imposition by the US in June 2018 of a 25 % tariff on steel and 10 % tariff on aluminium products. EUJEPA provides a major platform for both the EU and Japan to reinforce their commitment to rules-based trade and deepen their alliance against unilateral and protectionist policies. For Japan, the trade deal is an integral component of so-called third arrow (structural reforms) of 'Abenomics'. On 6 July 2018, Japan also ratified the Comprehensive and Progressive Agreement for Trans-Pacific Partnership (CPTPP-11), the successor to the TPP agreement, from which the US withdrew in January 2017. This agreement will unite 11 Asia-Pacific countries covering 13.5 % of global GDP[2] in a multilateral free trade zone. As well as Japan, Mexico and Singapore also ratified the agreement in April and July 2018 respectively. The agreement will enter into force 60 days after ratification by at least six of the eleven signatories. When implemented, the CPTPP-11 is expected to improve the competitiveness of Japan's economy by reducing the cost of sourcing intermediate inputs from partner countries. As a result, Japan's exports to the EU would increase as well (Felbermayr *et al*, 2018).

The impact of EUJEPA will also be affected by the United Kingdom's withdrawal from the EU in March 2019 in significant ways. Japan's total exports to the UK amounted to EUR 15.1 billion in 2017 (equivalent to 17 % of Japan's exports to the EU) and imports from the UK to EUR 16.8 billion (equivalent to 18 % of Japan's imports from the EU). Moreover, Japanese companies have invested heavily in the UK, as shown by their FDI position (with immediate ownership) of GBP 46.5 billion in 2016[3]. This investment strategy was mainly aimed at gaining access to the EU's single market. After the transition period for the UK leaving the EU ends in 2020, Brexit will reduce the market size for Japanese goods and services under the EUJEPA and affect

[1] The preferential access given by both the EU and Japan to various other trading partners has been a contributing factor to this reduction. For instance, the increase in EU trade with South Korea, following the implementation of the EUKFTA, has clearly had a negative impact. Another factor was the growth of China, which is now the EU's second largest export destination and largest source of imports.

[2] GDP figures for 2017 in current USD, World Development Indicators.

[3] Office of National Statistics, UK foreign direct investment, trends and analysis: July 2018.

the production operations of Japanese companies. On the EU side, the EUJEPA provides an additional source of external demand to partially counter the adverse effects of potential new trade frictions following Brexit, such as differences in tariffs, increase in customs procedures at the border and meeting rules-of-origin requirements.

In terms of historical timeline, the European Council gave the European Commission the mandate to start negotiations with Japan in November 2012. The first round of negotiations was held in April 2013. Following 18 rounds of negotiations, the parties reached a political agreement on 6 July 2017. The text of the agreement was finalised in December 2017 after which the Commission submitted it to the Council for approval in April 2018. The Council authorised the decision to sign the EPA and requested the consent of the European Parliament on 6 July 2018. The agreement was then signed on 17 July 2018 at the EU-Japan Summit in Tokyo. In terms of transparency, the negotiating directives, reports of negotiating rounds and finalised legal text were all published online by December 2017. Thematic factsheets, case studies of exporters and impact assessment reports were shared, and civil society dialogues and general meetings with the EU Trade Commissioner were held to increase public understanding and participation in the issues involved.

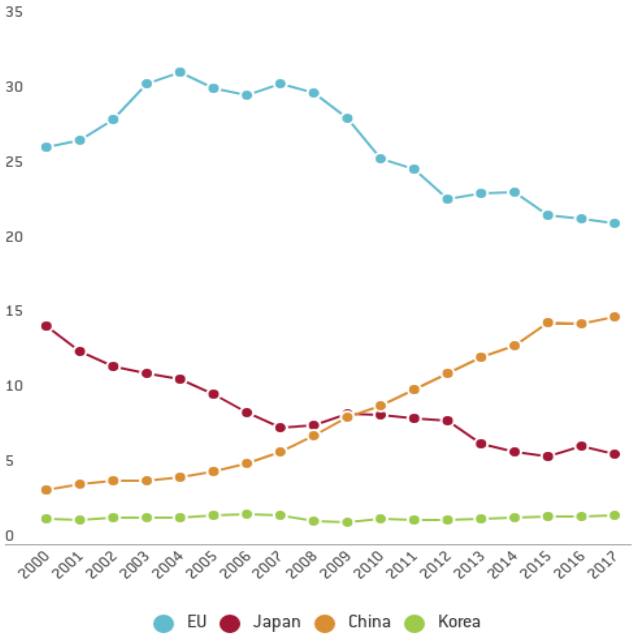
In advance of its final plenary vote on the agreement, the European Parliament therefore commissioned an independent Bruegel study on the topic. The study is structured as follows: Section II provides descriptive statistics on the trade and investment relationship between the EU and Japan. Section III gives a concise overview of the substantive provisions of the EUJEPA. In Section IV, legal texts from the family of the EU's new generation trade agreements (EUJEPA; the EU-Canada Comprehensive Economic and Trade Agreement, CETA; the EU-Singapore Free Trade Agreement, EUSFTA; the EU-South Korea Free Trade Agreement, EUKFTA) are compared in detail by studying WTO+ and WTO-X provisions. In Section IV, we use recent statistics to compare the EU's trade with Japan with the composition of EU trade with Canada, South Korea and Singapore. In Section V, we assess the economic and social impact of the agreement by reviewing the literature that utilises CGE and NQTT methods to estimate the impact of the EUJEPA. We highlight where these projections overlap or diverge with the actual outcomes observed after the implementation of the EUKFTA. Using detailed information on revealed comparative advantage and tariff schedules, we highlight potential areas of opportunities and threats for the EU. Finally, in Section VI we examine the framework for implementing the provisions of this trade deal and flag issues of concern. Section VII concludes our analysis with policy recommendations.

2 Descriptive statistics

2.1 Market size

The EU and Japanese economies together represent EUR 18 981 billion in GDP[4] and cover nearly 640 million inhabitants. GNI per capita is high at USD 32 778 for the EU and USD 38 550 for Japan in 2017 (current USD)[5]. At these levels, the EU's per-capita income is more than double and Japan's nearly three times the World Bank's threshold for classifying an economy as high income (>USD 12 235). Although China's economic size surpassed that of Japan in 2010 (Figure 1), Japan's per-capita private consumption at USD 26 745 (constant 2010 USD) was more than 14 times that of China's in 2016[6]. Hence, the EUJEPA connects two of the world's most highly advanced economies.

Figure 1. Share of World GDP, current USD (2000-2017)



***Source**: Bruegel based on World Development Indicators (2018), World Bank.*

2.2 Aggregate bilateral trade

Going by the latest figures from 2017, Japan was the EU's 7th largest trade partner with a 3.7 % share of the EU's overall import basket and 3.2 % share of its export basket. In aggregate terms, in recent years Europe has maintained a trade surplus with Japan that reached nearly EUR 6.5 billion in 2017 (Figure 2). A closer examination reveals that this aggregate trade surplus has been driven by the EU's exports of services to Japan, which stood at EUR 31 billion in 2016.

[4] USD 22 150 billion in current market prices in 2017, World Bank World Development Indicators.
[5] Average GNI per capita for high-income economies was USD 40 136 in 2017.
[6] World Development Indicators, 2018.

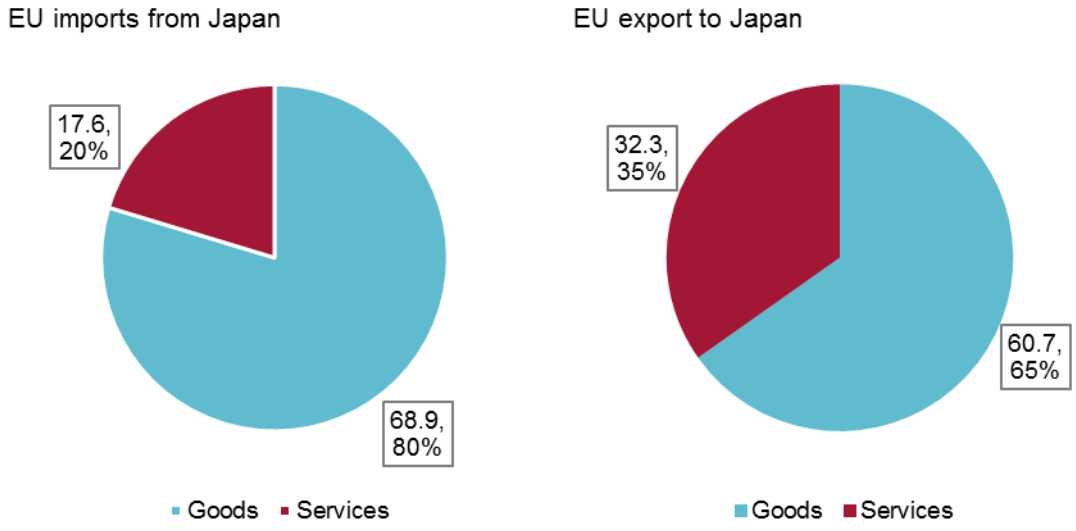
Figure 2. The EU's trade with Japan (EUR billion)

Note: Data for 2017 is provisional.
Source: *Eurostat.*

2.3 Trade in goods

Trade in goods makes up the greatest volume of bilateral trade between the EU and Japan on the export and import sides (Figure 3). The share of goods in value terms is even greater for the EU's imports (78.69 %) than its exports to Japan (65.16 %).

Figure 3. Composition of the EU's trade with Japan, 2016



Note: Services data is provisional.
Source: *Eurostat.*

Within the EU-28, in 2017 the top goods exporters to Japan were Germany, Italy, the UK and France, respectively accounting for 32.87 %, 10.82 %, 10.67 % and 10.53 % of the EU's EUR 60.66 billion of commodity exports to Japan in 2017. Germany, Italy, the UK and France were also the top four importers of Japanese goods, with respective shares of 23.28 %, 16.48 %, 14.37 % and 12.76 % of the EUR 68.89 billion worth of EU commodity imports from Japan in 2017.

Table 1 highlights the top product categories in the EU's goods trade with Japan. We can see how i) machinery and appliances, and ii) transport equipment are of major bilateral interest. Machinery and appliances accounted for 18.3 % of EU exports and 41.1 % of EU imports to and from Japan in 2017. Transport equipment was the second most traded product category, making up respectively 19.2 % and 8.9 % of the EU's total exports to and imports from Japan. Agricultural products have played a relatively smaller role up to now, comprising 5.4 % of EU exports to Japan in 2017. This could however change with the deepening of market access under the EUJEPA in the traditionally protected agri-food sector.

Table 1. Top products in the EU's trade with Japan, 2017

EXPORTS			IMPORTS		
Product	Value (Million Euros)	Share of total	Product	Value (Million Euros)	Share of total
Products of the chemical or allied industries	13,270	21.9	Machinery and appliances	28,342	41.1
Transport equipment	11,635	19.2	Transport equipment	17,402	25.3
Machinery and appliances	11,093	18.3	Products of the chemical or allied industries	6,098	8.9
Optical and photographic instruments, etc	5,720	9.5	Optical and photographic instruments, etc	5,559	8.1
Foodstuffs and tobacco	3,278	5.4	Plastics, rubber and articles thereof.	3,151	4.6

Source: *European Commission Trade Statistics.*
Note: Shares are calculated from total bilateral trade between EU and Japan in goods.

Japan is among the low-tariff countries for industrial goods with a trade-weighted tariff average of 1.4 % (Table 2). On the other hand, Japan's agricultural markets are relatively protected. Simple average applied most-favoured nation (MFN) tariffs stand at 13.3 % for agricultural goods, with high tariffs on animal products (10.6 %), dairy items (63.4 %), beverages and tobacco (15.1 %).

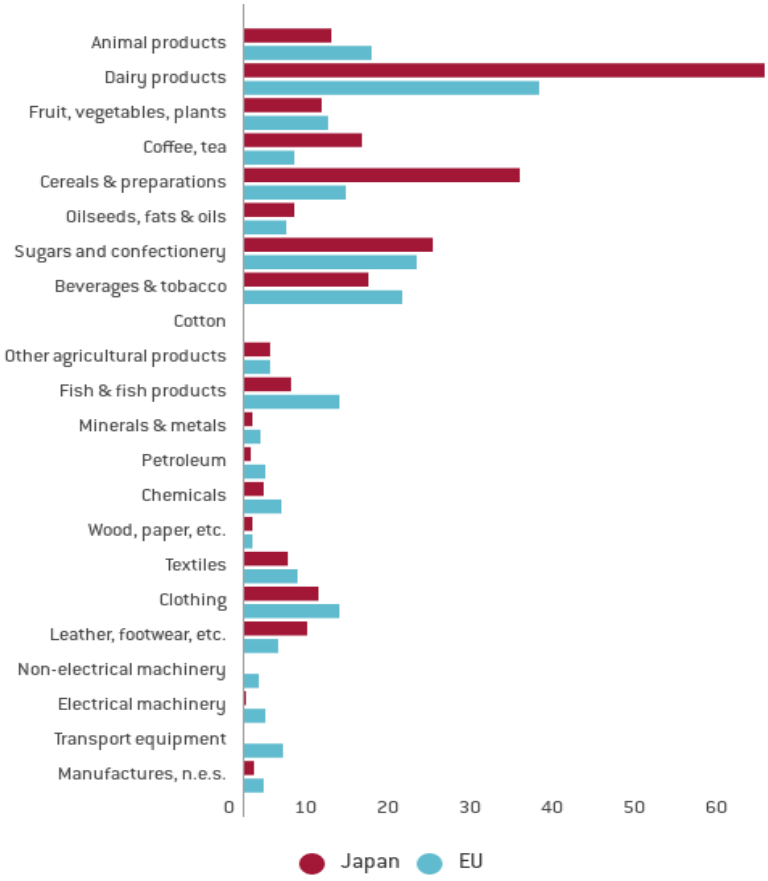
Table 2. Japan's tariff profile

Summary	Year	Total	Ag	Non-Ag
Simple average final bound		4.5	18	2.5
Simple average MFN applied	2017	4	13.3	2.5
Trade weighted average	2016	2.5	12.9	1.4

Source: *WTO Tariff Profile (2018).*

Figure 4 depicts the EU's and Japan's tariffs by harmonised system (HS) product categories. In terms of the average applied MFN tariffs, we note that Japanese tariffs are low across numerous sectors such as electrical machinery (0.1 %), transport equipment (0) and manufactures not elsewhere specified (n.e.s) (1.2 %). Prominent exceptions are clothing (9 %), leather and footwear (7.7 %).

Figure 4. Simple average of MFN applied duties, by HS categories for EU and Japan



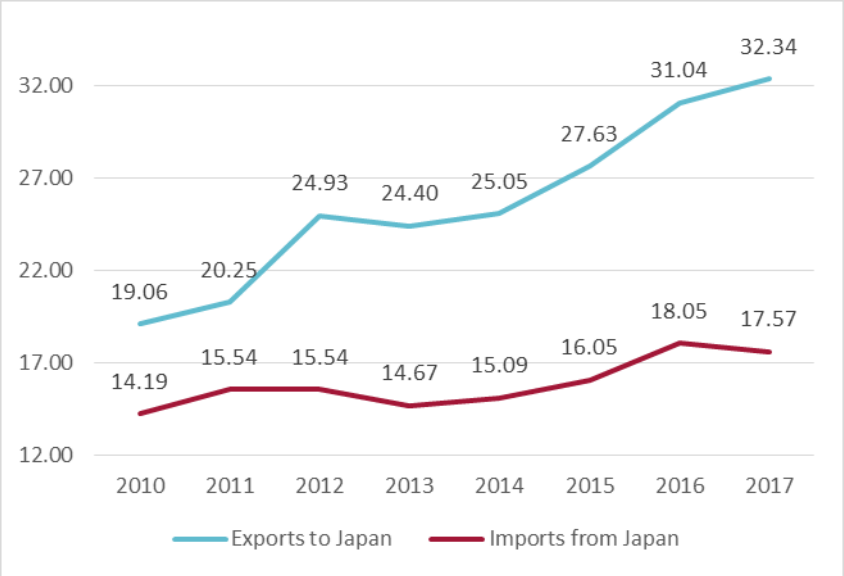
Source: *WTO Tariff Profile (2018).*

2.4 Trade in services

EU exports of services to Japan reached EUR 31.04 billion in 2017. In fact, the EU has maintained a significant trade surplus in services with Japan, which has been widening over time (Figure 5). The dominant sectors in the EU's total service exports to Japan in 2016 were financial services (23.53 %), telecommunications (14.5 %) and transport (13.97 %).

Financial services along with the telecommunication and information services sectors are clearly technologically sophisticated and data intensive, as evidenced by a high OECD index on digital 'maturity' in terms of investment in software and related inputs (OECD, 2017). This is in contrast to other services such as tourism and construction which are relatively low on this index of digital maturity. This is relevant as the EU and Japan concluded talks to recognise each other's personal data protection regimes as equivalent in July 2018. This mutual adequacy decision will be important in ensuring that the EU's new data protection rules under the General Data Protection Regulation (GDPR) do not disrupt the Union's service trade with Japan. The privacy dialogue was deliberately separated from EUJEPA by the EU. However, the successful conclusion of adequacy talks with Japan implies that the benefits of lowered non-tariff barriers (NTBs) in services sectors through EUJEPA can actually be realised, because transfer of data to Japan will be comparable to transfer of data within the EU.

Figure 5. EU's services trade with Japan (EUR billion)

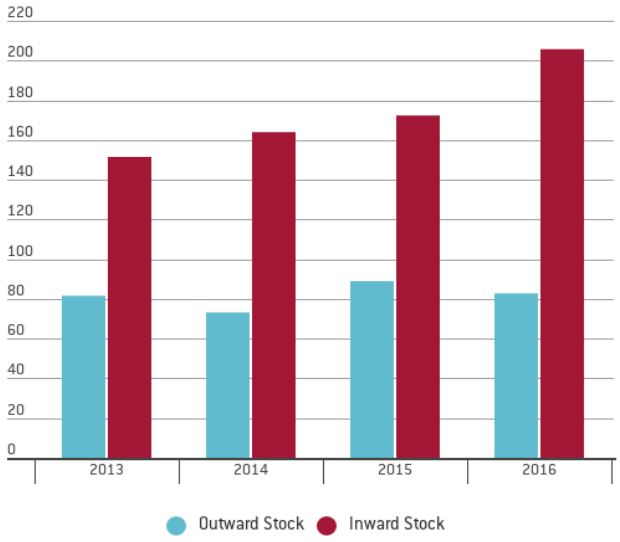


Note: 2017 data is provisional.
Source: *International trade in services (since 2010) (BPM6); own illustration.*

2.5 Bilateral investment

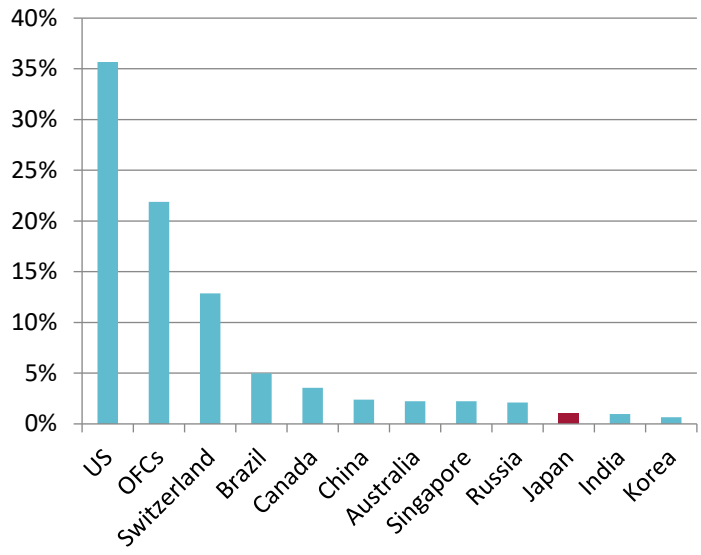
The EU's FDI stock in Japan has been stable since 2013 with an annual average growth rate of 0.5 %. By contrast, Japanese investments in the EU have increased substantially, from EUR 151 billion in 2013 to EUR 205 billion in 2016 (Figure 6). Japan accounted for only 1.1 % of the total extra-EU FDI in 2016 (Figure 7). By comparison, EU investments in the US and Canada are much higher, representing 38 % and 3.7 % of total extra-EU FDI. Japanese investment in the EU stood at USD 56.8 billion, 33.7 % of its total FDI stock abroad in 2017. However, it should be noted that a substantial share, approximately 38 %, of Japan's FDI stock in the EU is in the UK (JETRO, 2018; see Table 9 in Annex I).

Figure 6. EU FDI with Japan (EUR billion)



Source: *Eurostat Balance of Payments.*

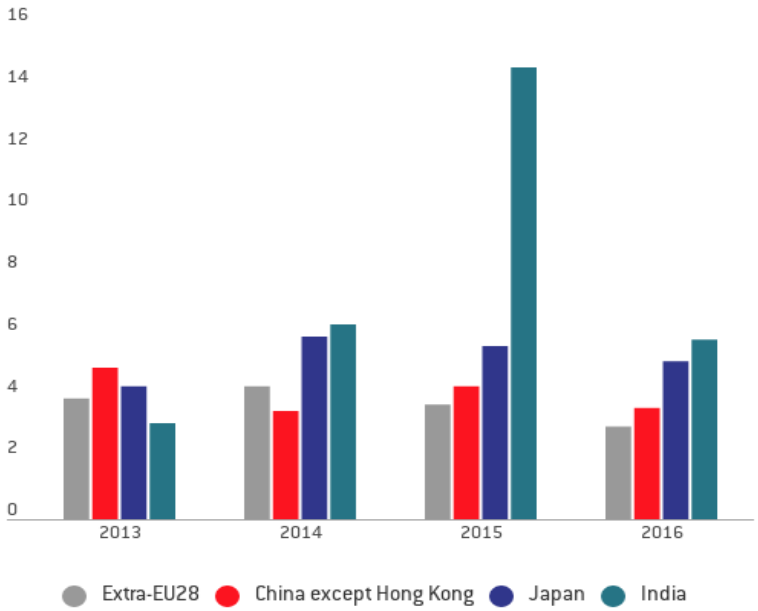
Figure 7. EU FDI positions by partner country, 2016 (% extra-EU total)



Source: *Eurostat Balance of Payments.*

Returns on investment in Japan are seen to be higher than the average extra-EU return since 2013. The return reached 5.1 % in 2016 for Japan (Figure 8). These statistics indicate that the Japanese market is lucrative for EU business capital.

Figure 8. Rate of return on EU direct investment (FDI income/FDI stocks)



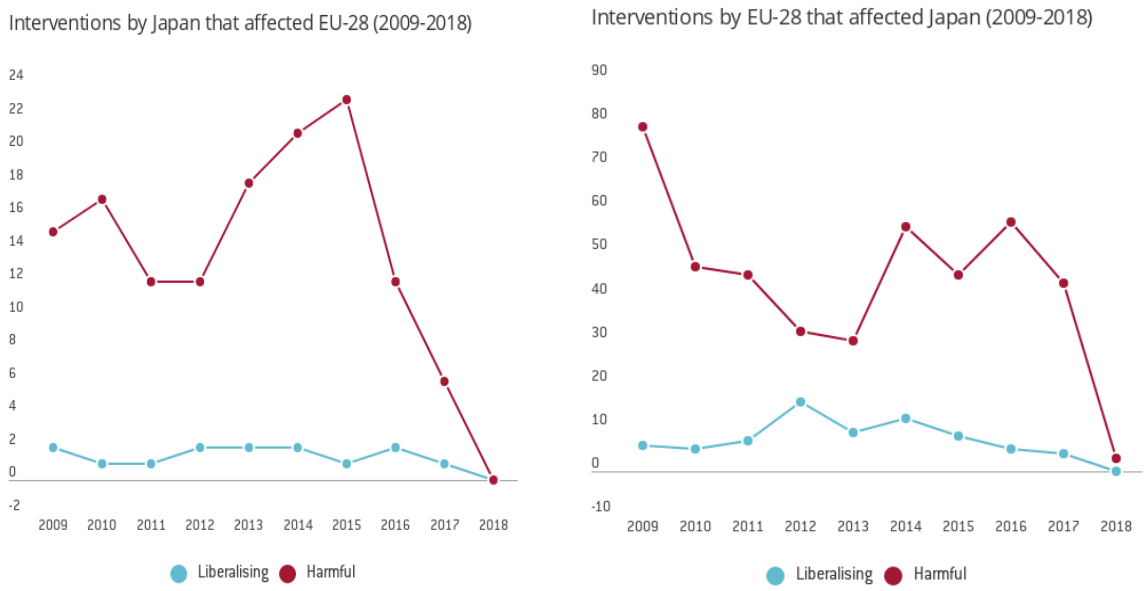
Source: *Eurostat Balance of Payments.*

2.6 Market openness

We first examine the number of liberalising and protectionist interventions between the EU and Japan over the period 2009-2018 using the Global Trade Alert Database[7]. Figure 9 shows that the number of harmful interventions by Japan against the EU exceeded liberalising interventions throughout this period, although there has been a reduction since 2015. Among the harmful interventions by Japan since 2009, the most affected sectors were motor vehicles and parts, engines, turbines, air and space transport services of freight. In terms of liberalising interventions, apparel and chemicals were amongst the main beneficiary industries. On the EU side, harmful interventions affecting Japan have been in financial, insurance and pension services, motor vehicles, transport services, special purpose machinery and its parts. Increased regulatory cooperation, tariff elimination in industrial products and continuous policy dialogue through the EUJEPA can ease these harmful interventions in the future.

[7] Each entry in the Global Trade Alert database documents government announcements concerning unilateral changes in the relative treatment of foreign versus domestic interests in trade, investment and labour force migration. According to whether an announced change results in a more equal or unequal treatment, the entry is classified as 'Liberalising' or 'Harmful'.

Figure 9. Interventions by Japan and EU-28 (2009-2018)



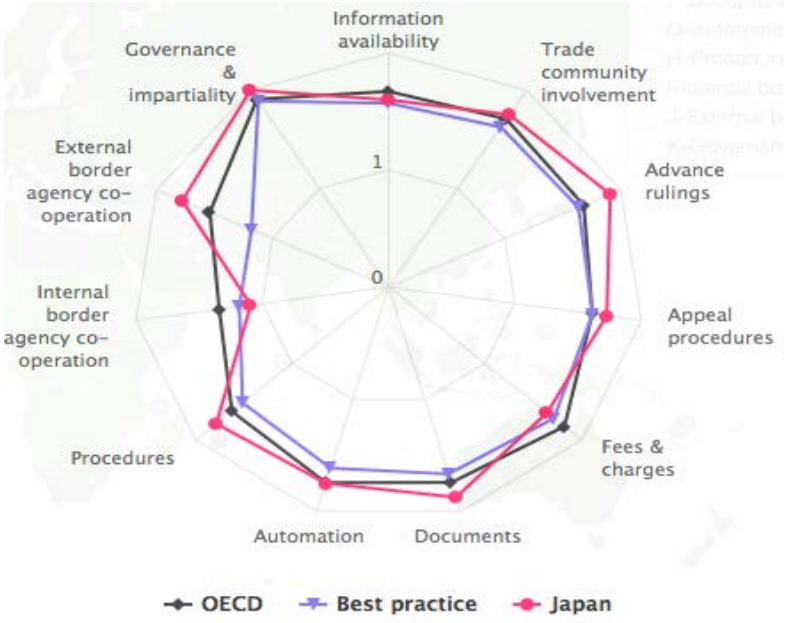
Source: *Global Trade Alert.*

In the service sector, Japan appears to be a relatively open country. Quantitatively, this can be seen in a lower score on the Services Trade Restrictiveness Index than the sample average in 21 out of 22 service sectors (2017). The restrictions are relatively high in legal services where there are limits on foreign entry, in courier services that have barriers to competition, air transport and telecommunications.

A low score (0.05) in OECD's FDI Regulatory Restrictiveness Index (2017), which measures statutory limitations on FDI, is indicative of the relative openness of the Japanese economy to foreign investors, as the OECD average is 0.07. On the other hand, the US (0.09), Canada (0.162) and China (0.316) are all relatively more restrictive as destinations for FDI. Except for a few countries in Europe (Austria, Poland and Sweden), all EU Member States are more open to FDI than Japan, with Luxembourg, Portugal and Slovenia being the least restrictive.

According to the OECD's Trade Facilitation indicators (OECD, 2018), Japan is close to best performance in all aspects (Figure 10). In the EU, several Member States including the Netherlands, France, Germany and Spain, perform better even than Japan in terms of the overall score. However, there is scope for additional reforms in the EU and Japan on the availability of information, especially through customs hotlines, and to improve formalities for documentation and procedures. The EUJEPA can provide an impetus for these reforms, as it carries commitments on increasing transparency of procedures and modernising customs rules to promote efficiency.

Figure 10. Trade Facilitation Indicator, Japan



Source: *OECD Trade Facilitation Indicators, 2017.*

3 Overview of commitments in EUJEPA

3.1 Provisions under the agreement

The EUJEPA is a highly comprehensive free trade agreement. In what follows, we detail in brief some of its main innovations.

Tariffs The EUJEPA was signed on 17 July 2018 by both parties. If it is ratified by the European Parliament and the Japanese Diet, the agreement will enter into force by March 2019, following which the trade clauses will begin to be implemented. Under the agreement, both parties would substantially liberalise bilateral trade in goods by reducing tariff barriers. Japan will eliminate 86 % of tariff lines on EU goods on entry into force and 97 % of tariff lines within 15 years. On its side, the EU will liberalise 96 % of tariff lines for Japanese goods on entry into force and 99 % of tariff lines by the end of the staging period. A more detailed discussion of staging periods is taken up in section 6.2.

Agricultural goods. For agri-food products, Japan will remove 85 % of tariff lines on EU exports over time (European Commission, 2018). This includes significant reductions in customs duties for the EU's major food exports to Japan such as pork (e.g. 4.3 % to 0 over 10 years for high value cuts), wine (15 % to 0 % on entry into force), beef (38.5 % to 9 % over 15 years), pasta and chocolates (complete tariff liberalisation in 10 years). For cheeses, the EUJEPA will deliver complete liberalisation for hard cheeses and provide tariff rate quotas (TRQs) with duty free access for fresh, processed and soft cheeses. TRQ volumes will gradually increase (from 20 000 to 31 000 tons) and in-quota tariffs will be phased out over 15 years. Certain sensitive products such as rice, seaweed and whale meat are excluded from the agreement.

Industrial goods. Both parties agreed to completely remove tariffs for these products over a transition period. Most notably, the EU will liberalise its automobile market following a transition period of seven years for Japanese cars and between entry into force and seven years for Japanese car parts. Japan will immediately eliminate duties for EU producers in sectors such as chemicals, textiles, apparel, metals, plastics and jewellery. Quotas will also be ended on entry into force for footwear, with gradual phase-out of high tariffs (e.g. 27 % to 0 % over 10 years for ski-boots).

By comparison, 70 % of tariff lines were duty-free on the provisional application of the EUKFTA. The applied trade-weighted average tariff on EU products was reduced from more than 8 % to 2 % in the first three years of the FTA (Civic Consulting and Ifo Institute, 2017). On the EU side, tariffs on Korean goods were also lowered significantly from a trade-weighted average of 2.5 % to 0.5 % in the post-FTA period. In the agri-goods industry, South Korea was traditionally protectionist with a trade-weighted tariff of 49 % in the pre-FTA period. By 2014 however, these had been reduced on a preferential basis for EU goods to 28 %. The EU also liberalised its agricultural markets for South Korea by reducing trade-weighted tariffs from 11 % to 3 %.

Non-Tariff Barriers The Japanese market has been low on tariffs but features significant formal and informal NTBs. The EUJEPA acts upon this, with multiple provisions to address divergent standards and technical requirements, cumbersome conformity assessment procedures and administrative issues that were highlighted in the Commission's Impact Assessment Report (CIAR) in 2012 (European Commission, 2012). Japan is aligning itself with international standards on medical devices (Quality Management Systems), textile labelling (ISO[8] international care labelling), motor vehicles (UNECE[9] international vehicle regulations) and pharmaceuticals (ICH[10]). This regulatory shift will promote EU exports by reducing the financial and administrative burden for firms, arising from dual testing and complex conformity assessment

[8] International Organization for Standardisation.
[9] United Nations Economic Commission for Europe.
[10] International Council for Harmonisation.

procedures. Under the EUJEPA, Japan will simplify approval and clearance procedures for sanitary and phytosanitary (SPS) measures and reform its notification system for pharmaceuticals, medical devices and cosmetics.

Common features with the EUJEPA can be found in the NTB reductions brought about by the EUKFTA. South Korea recognised UNECE as the relevant standard-setting body for motor vehicles and adopted an in-built mechanism to ensure continued alignment of standards with the EU in the future. The agreement also reduced the burden of third-party testing for EU electronics in South Korea and promoted policy coordination in SPS and TBT measures. Akin to the EUJEPA, government procurement was liberalised significantly. Going beyond the WTO's Government Procurement Agreement (GPA), the EUKFTA opened up build-operate-transfer (BOT) contracts for EU companies.

Services The EUJEPA seeks to promote bilateral trade in a broad range of services but does not require governments to deregulate or privatise the provision of public services such as healthcare, water supply and education. Here, we list a subset of provisions for certain priority service sectors. For cross-border services, the EUJEPA includes provisions on national treatment, MFN treatment and market access. In telecommunications, the agreement covers issues such as mobile roaming, number portability and confidentiality of users' traffic data. In financial services, the agreement calls for deeper regulatory cooperation and establishes a Joint Financial Regulatory Forum for this purpose. In e-commerce, the parties commit to keep electronic transmissions duty-free, recognise the legal validity of electronic contracts and signatures and may not require source codes to be transferred or accessed. In postal and courier services, the EUJEPA will attempt to build a level-playing field for EU suppliers and their main competitors such as Japan Post.

The EUJEPA also provides for the movement of natural persons for business services under mode 4 of the WTO's General Agreement on Trade in Services (GATS) including traditional categories such as intra-corporate transferees and contractual service suppliers, as well as two new profiles of short-term business visitors and investors that were committed in CETA. Similarly to CETA, the EUJEPA also includes a commitment to allow spouses and families to accompany service professionals on their temporary postings.

Investment The agreement contains provisions to promote and facilitate bilateral investment. The EUJEPA includes commitments on national and MFN treatment for enterprises of the other party. It also prohibits an extensive list of performance requirements (e.g. given level of domestic content, given value of exports/imports etc.) as conditions for establishing or operating enterprises in goods and service sectors. These provisions on investment liberalisation are complemented with guarantees on the free movement of capital, payments and transfers between the economies, though these can be suspended in the event of serious balance-of-payments crises or macroeconomic difficulties. As no prior investment agreement exists between any of the EU Member States and Japan, the provisions of the EUJEPA play an important role in establishing a fair and predictable investment climate between the partners.

However, outstanding issues on investment remain. The EU is currently engaged in negotiations with Japan on investment protection standards and the adoption of an Investment Court System (ICS). CETA was the first EU trade agreement to replace the earlier investor-state dispute settlement (ISDS) mechanism with the ICS. The ICS has also been incorporated recently into the EU-Singapore Investment Protection Agreement, the EU-Vietnam FTA (that is split to separate Investment Protection agreement) and the modernised trade pillar of the EU-Mexico Global Agreement. This shift in the EU's position is the result of numerous concerns about the ISDS that were also raised in the public consultation on investment protection and ISDS launched by the European Commission in 2014. The key advantages of the ICS over ISDS are i) increased transparency: through public hearings, online availability of case documents and third-party submissions by NGOs, consumer groups and so on; ii) increased legal certainty: due to the establishment of a permanent and institutionalised tribunal; and iii) increased protection of governments'

right to regulate: through limitations of the grounds on which investors can challenge states. However, Japan's position is to discuss reforms of the ISDS through the United Nations Commission on International Trade Law (UNCITRAL) Working Group III. A factor explaining the Japanese government's preference for ISDS could be that it has not yet faced investor-state arbitration and only three publicly known ISDS cases have been initiated by Japanese investors[11].

Public Procurement The EUJEPA provisions go further than commitments by the EU and Japan in the WTO's Government Procurement Agreement (GPA). Japan has agreed to provide non-discriminatory access to EU firms in the procurement markets of: i) 48 'core' cities (300 000 inhabitants each); ii) new categories of services beyond Annex 5 of the GPA such as telecommunications-related and credit-reporting services; and iii) additional government entities such as the Information Technology Promotion Agency and Pharmaceutical and Medical Devices Agency. Japan will also ease qualifying conditions for EU suppliers, by recognising accomplishments in the company's home country as equivalent. For instance, EU suppliers are required to undergo a Business Evaluation (or 'Keishin') under the Construction Business Law of Japan before they can submit tenders for public construction contracts. Under the EUJEPA, Japan has committed to ensuring that EU firms are not discriminated against in these compliance procedures and, wherever appropriate, supplier credentials[12] realised outside of Japan would be recognised as equivalent. Furthermore, the EU has secured improved access to Japan's markets for infrastructure and equipment as Japan is set to remove the Operational Safety Clause in the railways sector.

Both parties have made an important commitment to publish notices of intended and planned procurement on a single portal on the internet. This provision is aimed at increasing the availability of information and facilitating entry of new suppliers into public procurement markets. The EU and Japan will also recognise test results from conformity assessment procedures conducted by the other party, further reducing the administrative costs associated with compliance.

Intellectual Property Chapter 14 of the EUJEPA contains general provisions and provisions relating to standards and enforcement procedures on intellectual property. Under general provisions, parties reaffirm their obligations under the WTO Agreement on Trade-Related Aspects of Intellectual Property Rights (TRIPS Agreement), agree to national and MFN treatment in IPR protection and promotion of transparency in the administration of IP systems. The agreement then sets out provisions on protection of copyrights, trademarks, patents, industrial designs, trade secrets, etc, and civil enforcement of IP including border measures for goods that are exported and imported. A key commitment under the EUJEPA relating to IP is Japan's recognition of GIs for 205 EU agri-food products, with the possibility of adding new GIs to the protected list.

Competition Policy Chapter 11 of the agreement sets out principles and commitments on fair and free competition in trade and investment relations between the parties. Each party is required to maintain its competition law to address potential abuse of dominant market positions, anti-competitive mergers, etc. Furthermore, the agreement envisages cooperation between competition authorities for the development of policies and their effective enforcement. This chapter is carved out of dispute settlement.

Trade and Sustainable Development (TSD) The TSD chapter preserves parties' right to regulate and determine their own domestic policies to protect human, animal and plant health. The chapter calls for upholding existing standards for protecting consumers, labour rights and the environment while preventing the parties from waiving these protections in order to promote trade and investment. In the case of labour standards, Article 16.3 of the EUJEPA contains obligations for parties to make sustained

[11] Interviews with public and private sector actors and business representatives confirmed a certain degree of concern on the Japanese side, and a desire to reach an agreement on the ISDS in the future.
[12] These credentials may include amount of equity capital, sales for completed construction work, number of operating years in the business, number of technical staff, etc.

efforts to ratify fundamental ILO conventions. Japan has ratified six of the eight fundamental conventions and hence must make progress on ratification of the remaining two, which concern the abolition of forced labour (C105) and discrimination in employment and occupation (C111).

In a first for an EU FTA, both parties also reaffirm their obligations to **the United Nations Framework Convention on Climate Change (UNFCCC) and the Paris agreement** on climate change. Recognising the importance of biodiversity, the agreement calls for sustainable use of natural resources; combating illegal trade in endangered species and timber; implementing effective measures against illegal, unreported and unregulated fishing; and exchanging information on best practices. The relevant chapter suggests further cooperation between parties in numerous areas such as labelling schemes to promote sustainable trade, corporate social responsibility and climate-friendly technologies. Civil society organisations can offer their views on socio-economic and environmental issues relating to the EUJEPA through domestic advisory groups within each party and a new Joint Dialogue with civil society. Disputes relating to this chapter will be discussed through government consultations, with recourse to an independent panel of experts.

Small and Medium-sized Enterprises (SMEs) Chapter 20 of the agreement envisages cooperation between parties on increasing awareness amongst SMEs on a wide range of issues spanning customs procedures, rules of origin, technical requirements, conformity assessments and business registration. Specifically designated SME Contact Points will foster policy coordination between parties and highlight issues relevant to SMEs in the implementation period. This is an important step towards removing information barriers that limit the ability of SMEs from exporting to the parties. Increased transparency and availability of information in English can support SMEs in overcoming language differences, bureaucratic red tape and related non-tariff barriers.

Corporate Governance The EUJEPA is the first international trade agreement to include provisions on corporate governance. Parties are required to develop an effective corporate governance framework that protects and facilitates shareholders' rights and promotes information disclosure on issues such as the company's capital structure and shareholdings. The agreement should encourage responsible decision-making by boards and provide rules for fair and transparent takeovers. These new commitments were made with the aim to: i) set high standards in corporate governance; ii) reduce behind-the-border barriers on investment arising from diverging regulations on the management of firms; and iii) increase investor confidence, investment and competitiveness.

Regulatory Cooperation The EUJEPA is the first EU trade agreement to include a separate chapter on regulatory cooperation. It is forward-looking, as it sets up a joint body called the Regulatory Cooperation Committee (RCC) to identify areas of future cooperation between parties. The RCC is intended to enhance bilateral cooperation on international standard setting and facilitate the exchange of information on best practices and planned regulatory measures. Regulatory authorities are encouraged to adopt good practices including public consultations, impact assessments and periodic retrospective evaluation of measures in force. These provisions seek to promote an *'effective, transparent and predictable regulatory environment'* to encourage trade and investment flows. This cooperation is purely on a voluntary basis and its scope is limited to EU regulatory measures and the European Commission as the relevant regulatory authority for the EU. Hence, the provisions do not apply to EU Member States and preserve their right to regulate in matters relating to public health, the environment, financial stability and energy security, among others.

3.2 Opportunities in new technologies

Japan and the EU are leaders in science and technology. Japan spends 3.1 % of its GDP on R&D and 60.1 % of its population in the 25-34 age group has completed tertiary education (OECD, 2016). Moreover, the

country outperformed the EU in the 2017 European Innovation Scoreboard, showcasing its advantages in business R&D, innovation collaboration and IP applications.

This trade deal will strengthen cooperation between EU and Japan on innovation, supporting the existing Agreement on S&T Cooperation which came into force in 2011, and the new EU-Japan Strategic Partnership in Research and Innovation, which was endorsed at the 2015 EU-Japan Summit. Exchange of best practices, technologies and researchers between the EU and Japan can yield innovative solutions to common challenges including climate change, ageing populations and energy security.

First, the adoption of common technical standards and simplification of approval procedures in pharmaceuticals and medical devices will encourage entry and investment by firms in the life-sciences industry. The interaction between Japan's demand for healthcare[13] and scientific leadership can incentivise innovation by EU firms in this sector.

Second, the tariff cuts for Japanese cars and car parts in the EU can incentivise further investment and exchange of technologies in the motor vehicle industry. The agreement's Automotive Annex has resolved several NTB concerns, allowing EU type-approved vehicles to be sold in Japan without modifications, and foresees joint work to set international standards in the future. The removal of such tariff and regulatory barriers can promote investment in developing advanced technologies spanning electric cars, fuel-efficient engines and autonomous driving.

Another key area of mutual interest is information and communications technology (ICT). Japan is one of the least restrictive economies in the world according to ECIPE's Digital Trade Restrictiveness Index Report (Ferracane *et al*, 2018) and is a signatory to the WTO's Information Technology Agreement. In July 2018, the EU and Japan also concluded talks on reciprocal recognition of each other's adequacy in terms of data protection and the European Commission has started the EU process to adopt this adequacy decision under the GDPR. This would clear the path for the EU and Japan to establish a leading zone for safe data transfers and could magnify the benefits from EUJEPA in digital economy sectors that are data-intensive. Furthermore, the EUJEPA allows for non-disclosure protection on source code, opens up services markets and prevents performance requirements for covered enterprises. By establishing frictionless and safe cross-border data flows, the EU and Japan can bolster productive activities in e-commerce, 5G telecommunications and artificial intelligence-related applications.

Better protection of IPR in Japan and clearer provisions on the same in EUJEPA will ensure that firms operate under a stable legal framework and in a secure innovation environment. This regulatory certainty is crucial in promoting innovative activity by firms.

[13] Health spending in Japan was 10.7 % of GDP in 2017, the sixth highest amongst OECD economies. See Health Expenditure and Financing, OECD Statistics.

4 Comparison of EUJEPA with other new generation FTAs

4.1 Coverage of issues

The EUJEPA reflects Europe's ongoing efforts to develop models for a new generation of trade agreements. Unlike earlier agreements, these new agreements are more ambitious in scope, with provisions that extend beyond tariff reductions and trade in goods. To understand the EUJEPA's position in the taxonomy of new generation FTAs, we study its legal text and compare its scope to the CETA, EU-Singapore FTA (EUSFTA) and EU-South Korea FTA (EUKFTA). This joint analysis of treaty design is essential to assess the projected impact of the EUJEPA. However, it is beyond the scope of this study to undertake an issue-by-issue comparison of provisions under the EUJEPA with CETA, EUSFTA and EUKFTA. Hence, our analysis is structured as follows.

First, we highlight the common elements and differences in issues covered by these trade agreements using the methodology proposed by Horn, Mavroidis and Sapir (2010). This study classified policy areas in FTAs as either WTO+ (building on existing WTO commitments) or WTO-X (issues beyond WTO's current scope) and then coded them according to their legal enforceability. Identification of the legal enforceability of provisions is based on their inclusion in the dispute settlement procedures as defined by the FTA, clearly specified obligations and the use of language that is precise or includes commitments (e.g. 'shall cooperate', 'shall not fail to', 'neither party may impose'). This methodology has been used extensively in the literature, most prominently by the World Bank's Hofmann *et al* (2017) to study the content of 279 FTAs. Tables 3 and 4 give an overview of the four FTAs (EUJEPA, CETA, EUSFTA and EUKFTA), with AC and LE indicating 'area covered' and 'legal enforceability' respectively. AC is coded as 0 (not covered) or 1 (covered) and LE is coded as 0 (not legally enforceable), 0.5 (legally enforceable language but carve-out from dispute settlement) or 1 (legally enforceable subject to dispute settlement).

While this analysis offers a concise comparison of treaty scope, it does not uncover the variety of commitments made on each issue covered by these FTAs. Therefore, the second step in our analysis is to undertake more detailed comparisons in policy areas where this variety is substantial and examinable, given the priorities and limitations of this study. With these two criteria, we compare provisions on state aid and public procurement that are WTO+ areas and on trade and sustainable development as a WTO-X area.

WTO+ comparison

All four FTAs have a very high degree of coverage of WTO+ policy areas with commitments on services trade, technical barriers, public procurement and intellectual property. They all cover trade in goods as well as trade in services, reflecting the importance of service inputs for production and export of goods. The majority of these provisions are legally enforceable, with the exception of SPS, anti-dumping and countervailing measures that are not subject to dispute settlement. Hence, in comparing coverage of WTO+ issues, the EUJEPA appears very similar to the EU's other three new generation FTAs. Collectively, these four agreements differ in several significant ways from earlier EU agreements, such as with South Africa, CARIFORUM and Israel (analysed in Horn, Mavroidis and Sapir, 2010). Unlike their predecessors, the new generation agreements cover new policy areas, specifically, trade related investment measures (local content requirements, export performance of FDI) and the elimination of export taxes. We now delve into more detailed comparisons of the commitments in two of these WTO+ areas.

State aid Commitments on government subsidies build on the WTO Agreement on Subsidies and Countervailing Measures. In the EUJEPA and EUSFTA, each party is requested to notify the other of the legal basis, form, amount and, if possible, name of the recipient of subsidies, every two years from entry into force. Such transparency provisions are similar in CETA, with the exception of recipient details which are

not mentioned in that agreement. The notification procedures are slightly different in EUKFTA, with parties having to report annually on subsidies that might affect international trade.

EUJEPA, EUSFTA and EUKFTA prohibit **government subsidies** to guarantee the debts or liabilities of an enterprise without imposing limits on the amount and duration, or to restructure an insolvent enterprise without requiring it to prepare a credible restructuring plan. In case a state subsidy is considered to have a significant negative impact on trade and investment interests, parties can request a consultation in EUJEPA and CETA. In EUJEPA, the requested party has to provide the relevant information (or an explanation if it cannot provide) within 90 days of receiving the request. In EUSFTA and EUKFTA, parties commit to reviewing progress in the implementation of provisions relating to state aid every two years following entry into force of the agreement. None of the FTAs include state subsidies in the ambit of the dispute settlement mechanism. Under EUSFTA, parties also agreed to hold a dialogue within two years of entry into force on developing rules applicable to subsidies.

Public Procurement All four trade agreements reviewed contain dedicated chapters on public procurement markets. These commitments extend beyond the WTO's GPA and are legally enforceable under the dispute settlement procedures of the agreements. EUJEPA, CETA and EUSFTA include a requirement for procurement entities to evaluate the commercial and technical abilities of a supplier based on its activities both inside and outside the territory. Such provisions address the issue of discrimination through technical specifications and encourage competition. The EUJEPA goes further, with additional clauses on the mutual recognition of test reports for suppliers. EU suppliers can also bid for contracts with governments at the sub-central level in EUJEPA and CETA. This is not applicable to Singapore, which has no sub-central governments. Rules on procurement procedures, information and transparency are derived from the WTO's GPA. However, EUJEPA, CETA and EUSFTA incorporate an additional commitment to set up a single electronic portal for accessing all notices pertaining to public procurement. In terms of implementation, a stand-alone Committee on Government Procurement is established under the EUJEPA and CETA. In comparison, issues relating to procurement are referred to a Committee on Trade in Services, Investment and Government Procurement in the EUSFTA and to the Government Procurement Working Group in EUKFTA.

Table 3. Classification of WTO+ areas

Provisions/FTA	EUJEPA		CETA		EUKFTA		EUSFTA	
	AC	LE	AC	LE	AC	LE	AC	LE
FTA Agr	1	1	1	1	1	1	1	1
FTA Ind	1	1	1	1	1	1	1	1
Customs admin	1	1	1	1	1	1	1	1
Exp. Tax	1	1	1	0	1	1	1	1
SPS	1	1	1	1	1	0.5	1	1
TBT	1	1	1	1	1	1	1	1
STE	1	1	1	1	1	1	1	1
AD	1	0.5	1	0.5	1	0.5	1	0.5
Subsidies & CVM	1	0.5	1	0.5	1	0.5	1	0.5
State Aid	1	1	1	1	1	1	1	1
Pub.Proc (GPA)	1	1	1	1	1	1	1	1
TRIMs	1	1	1	1	0	0	1	1
GATS	1	1	1	1	1	1	1	1
TRIPs	1	1	1	1	1	1	1	1

Source: *Bruegel.*
Note: AC: area covered; LE: legally enforceable.

WTO-X Comparison

We can identify further areas of overlap and divergence by studying WTO-X issues. First, all four FTAs include aspects relating to environmental law and labour market regulations. Hence, their objective is not limited to expanding market access but extends to promoting equitable and sustainable trade. Second, there is considerable overlap in terms of competition policy as this area is covered but carved out from dispute settlement in all four agreements. Investment promotion (national treatment, exchange of information, simplification of procedures etc) is another important WTO-X issue that is covered and legally enforceable in the four FTAs. Despite the similarities in coverage of investment promotion, there are significant differences in terms of investment protection and dispute resolution across the four FTAs. Negotiations on investor-state dispute settlement are ongoing between the EU and Japan, whereas CETA already includes the new Investment Court System supported by the EU. South Korea is in discussions with the European Union regarding this Investment Court System. EUKFTA does not currently cover investor protection as negotiations had started on the agreement before FDI became an exclusive competence of the EU under the Treaty of Lisbon, which entered into force in December 2009. In the case of Singapore, the terminology contained in the separate Investment Protection Agreement (EUSIPA) is that of an 'investment tribunal'.

Except for EUKFTA, the other three agreements (EUJEPA, CETA and EUSFTA) include cooperation on research, technologies and energy-related issues. The EUJEPA also includes additional WTO-X areas. It contains a separate chapter on corporate governance (Chapter 15) that encourages the development of an effective framework to protect shareholders' rights, promote information disclosure and responsible board decision-making. This is a novel dimension of bilateral cooperation that was not present in earlier EU FTAs. The EUJEPA and CETA also include SME-specific provisions (e.g. information sharing).

Table 4. Classification of WTO-X areas

Provisions/FTA	EUJEPA		CETA		EUKFTA		EUSFTA	
	AC	LE	AC	LE	AC	LE	AC	LE
Anti-Corruption	0	0	0	0	0	0	0	0
Competition Policy	1	0.5	1	0.5	1	0.5	1	0.5
Environmental Laws	1	0.5	1	0.5	1	0.5	1	0.5
IPR	1	1	1	1	1	1	1	1
Investment	1	1	1	1	1	1	1	1
Labour Market Regs	1	0.5	1	0.5	1	0.5	1	0.5
Movement of Capital	1	1	1	0	1	1	1	0
Consumer Protection	0	0	0	0	0	0	0	0
Data Protection	0	0	0	0	0	0	0	0
Agriculture	1	0	1	0	0	0	0	0
Approx of Legislation	0	0	0	0	0	0	0	0
Audiovisual	0	0	0	0	1	0	0	0
Civil Protection	0	0	0	0	0	0	0	0
Innovation Policies	0	0	0	0	0	0	0	0
Cultural Cooperation	0	0	0	0	1	0.5	0	0
Econ Policy Dialogue	0	0	0	0	0	0	0	0
Education & Training	1	0	1	0	0	0	1	0
Energy	1	0	1	0	0	0	1	1
Financial Assistance	0	0	0	0	0	0	0	0
Health	0	0	1	0	1	0	0	0
Human Rights	1	0	1	0	0	0	0	0
Illegal Immigration	0	0	0	0	0	0	0	0
Illicit Drugs	0	0	0	0	0	0	0	0
Industrial Cooperation	0	0	0	0	0	0	0	0
Information Society	1	0	1	0	1	0	1	0
Mining	0	0	0	0	0	0	0	0
Money Laundering	0	0	0	0	0	0	0	0
Nuclear Safety	0	0	0	0	0	0	0	0
Political Dialogue	0	0	0	0	0	0	0	0
Public Administration	0	0	0	0	0	0	0	0
Regional Cooperation	1	0	1	0	1	0	1	0
Research and Technology	1	0	1	0	0	0	1	0
SME	1	0	1	0	0	0	0	0
Social Matters	0	0	0	0	0	0	0	0
Statistics	0	0	0	0	0	0	0	0
Taxation	0	0	0	0	0	0	0	0
Terrorism	0	0	0	0	0	0	0	0
Visa & Asylum	0	0	0	0	0	0	0	0
Corporate Governance	1	0.5	0	0	0	0	0	0

Source: *Bruegel.*
Note: AC: area covered; LE: legally enforceable.

Analysing the legal enforceability of provisions, we observe that the majority of WTO-X areas in these EU FTAs are not legally binding. The key policy areas for which the agreements contain legally-binding language subject to dispute settlement are: i) competition policy; ii) intellectual property rights; iii) investment; and iv) movement of capital. Finally, some policy areas have been broadly omitted from the EUJEPA and from other recent EU agreements, but were covered by earlier EU agreements. These include issues related to national security such as nuclear safety, illegal immigration, terrorism and civil protection that are excluded from all four recent EU FTAs. On the other hand, national security issues are covered in the Strategic Partnership Agreements between the EU and Japan and Canada.

The EUJEPA is different from other new generation EU FTAs in another key way. It is an 'EU-only' agreement, covering only issues that are under the exclusive competence of the EU. As such, the EUJEPA only requires approval from the Council and ratification by the European Parliament. Following the opinion of the EU Court of Justice (May 2017), the EUSFTA will be 'EU-only' whereas the EUSIPA will need ratification by individual Member States. In contrast, CETA and EUKFTA were 'mixed' agreements requiring ratification by

Member States' national parliaments, because they covered issues that are under EU's exclusive and shared competences.

We now provide a more detailed comparison of trade and sustainable development commitments contained in the four FTAs. In keeping with the EU's 'Trade for All' strategy, all agreements include commitments to protect high social and environmental standards. They all recognise that each party has the right to determine its own sustainable development policies but parties should not weaken or waive protections for the environment and labour in order to encourage trade and investment.

All four agreements stress the importance of multilateral environmental agreements. However the EUJEPA is the first EU FTA to include commitments related to **the Paris Agreement** (December 2015)[14]. Further to this, EUJEPA, CETA and EUSFTA include clauses that **encourage effective implementation of measures on sustainable forest management, conservation of biological diversity, combating illegal logging and IUU fishing.** However, whaling remains an outstanding issue with Japan as the EU has banned trade in whale products. Although this issue is not explicitly mentioned in the text of the EUJEPA, the EU intends to use the agreement as an additional platform to foster dialogue with Japan on this subject.

With respect to labour, all four agreements contain commitments on promoting trade that is conducive to full, productive and decent work. **They reaffirm obligations deriving from ILO membership and the continuation of efforts to pursue ratification of fundamental ILO Conventions** and other ILO Conventions considered appropriate by the party. CETA has additional clauses that relate to spreading public awareness of labour laws and developing a preventive culture of health and safety. The EUJEPA and CETA also reaffirm parties' commitments to the principles enshrined in the Universal Declaration of Human Rights. These areas are not explicitly mentioned in the EUSFTA and the EUKFTA.

Another common feature of the four agreements is a commitment to monitor, assess and review the impact of each agreement on sustainable development. In addition to this, the EUSFTA further promotes an exchange of views on methodologies and indicators for conducting such trade sustainability impact assessments**.** All four agreements encourage cooperation in **promoting corporate social responsibility**, **green technologies, eco-labelling schemes and exchange of information on best practices**. In this list, CETA also explicitly includes the promotion of life-cycle management of goods, including carbon accounting and extended producer-responsibility for the treatment and disposal of consumed goods. Dialogues with civil society organisations (e.g. Civil Society Forums, Joint Dialogue with Civil Society) and domestic advisory groups are established in all the agreements in order to gather views and obtain recommendations on the sustainable development aspects of each agreement. In case of disagreement, the common mechanism in all agreements is to request government consultations with the possibility of establishing a panel of experts to further examine the matter.

All in all, our analysis shows that the EUJEPA is closely aligned with CETA, EUSFTA and EUKFTA in terms of issue coverage and legal enforceability of WTO+ and trade-related WTO-X areas. Therefore, the EUJEPA can be considered as belonging to the family of 'new generation' EU FTAs, although it is not identical to the other three agreements. By introducing new provisions, such as those on corporate governance and climate change, the EUJEPA represents a further deepening of the EU's trade treaties.

4.2 Comparisons of bilateral trade patterns

Moving beyond our legal text analysis that benchmarks the EUJEPA relative to other trade deals, in this section we aim to illustrate the underlying trade patterns between the EU and the comparison countries we have considered (Japan, South Korea, Singapore and Canada). Figure 11 shows the composition of the EU's imports from Japan, South Korea, Singapore and Canada, grouped by SITC1 classifications.

[14] By the Conference of the Parties to the UNFCCC at its 21st session.

South Korea appears as the most similar comparator to Japan. From both countries the EU imports heavy machinery and transport equipment (in light blue), and in particular road vehicles and electrical machinery, while Singapore imports significant shares of chemicals and pharmaceutical products. In terms of imports from Canada, (precious) metals, and gold in particular, take the greatest share – something sharply at odds with the European import pattern in relation to Japan.

Figure 11. EU imports from Japan (top left), South Korea (top right), Singapore (bottom left), and Canada (bottom right)

***Source**: Eurostat Comext.*
Note: Data for 2017 by SITC2. SITC1 products are grouped by the same colour.

While Figure 11 offers the benefit of a quick comparison of trade patterns, we also carried out the analysis in a more granular and formal fashion. We therefore developed an index of bilateral trade flows between the EU and Japan, South Korea, Canada and Singapore. Intuitively, it is a similarity index comparing the

vector of imports[15] at the HS 2-digit product level[16]. The index can take on values between 0 and 1, where 1 implies that the product basket is identical to the trade relationship between the EU and Japan and 0 implies there is no overlap in product lines. Table 5 illustrates quantitatively what Figure 11 already suggests: that South Korean exports to the EU are those most similar to Japan's exports to the EU. Singapore is less similar to Japan and Canada is least similar. Table 5 shows the similarity index for the years 2015-2017, to reassure the reader that this relationship is stable, and not a consequence of one-offs or irregular trade data.

Table 5. Similarity index based on EU imports and selected trade partners

year	tradeflow	reporter	JPN	KOR	SGP	CAN
2015	Import	EU-28	1	0.722	0.539	0.316
2016	Import	EU-28	1	0.733	0.513	0.333
2017	Import	EU-28	1	0.707	0.455	0.305

Source: *Bruegel based on Eurostat Comext.*
Note: Similarity index computed as the Euclidean distance between the import vector of the EU from Japan and selected trade partners. 1 implies identical trade relationship to EU-Japan.

The main takeaway from this section is that EU imports from South Korea are the most similar to EU imports from Japan. This quantitative finding complements the data presented in section 4.1, which showed how the EUJEPA has the greatest level of similarity to the EU-South Korea FTA. All in all, these results will be relevant when trying to predict the likely impact of the EUJEPA (section 5).

[15] The reason why the pattern of EU imports from the four countries are considered, rather than EU exports to the four countries, is that, given at global level the EU holds a comparative advantage in a specific set of products/industries, there will be a smaller level of heterogeneity across all four destinations. Indeed, quantitatively, we observe that the index similarity at EU export level in 2017 is 0.780 for KOR, 0.764 for CAN and 0.574 for SGP. Nonetheless, even from this standpoint, Korea appears to the country most similar to Japan in terms of its trade pattern with the EU.

[16] More formally, the index is computed as the Euclidean distance between vectors, or else: $1 - (\sum i \in I \, |\text{EU-JP}i, - \text{EU-CC}i,|)/2$ where I is the set of all HS-5 product lines, and EU-CC*i*, is the share of product i in EU total imports from country CC, at time t.

5 Projected impact of EUJEPA

5.1 Literature review of economic impact

We reviewed a number of existing studies to provide an overview of the projected impact of the EUJEPA. These studies employ Computable General Equilibrium (CGE) and New Quantitative Trade Theory (NQTT) models to determine the causal impact of the agreement, with varying baseline assumptions (e.g. regarding FTAs in force) and multiple liberalisation scenarios, ranging from conservative to ambitious trade-cost reductions. Table 10 (Annex II) provides a clear mapping of the main model assumptions. General equilibrium outcomes are reported in Table 11 (Annex II). The main focus will be on projections reported in the latest studies such as DG Trade (European Commission, 2018) and Ifo Working Paper (Felbermayr *et al*, 2018). The earliest studies such as SNBT (Swedish National Board of Trade, 2009) and Ecorys (Plaisier *et al*, 2009) may not be as relevant given the vintages of data used for setting baseline scenarios and assumptions of the outcome of EU-Japan negotiations on trade and investment. Nonetheless, we report the assessment of these studies in order to indicate the shifts in expected outcomes.

Finally, we compare the results of the aforementioned studies on EUJEPA with those on EUKFTA (Tables 12 and 13, Annex III). Of these, two are ex-ante CGE evaluations of EUKFTA. To understand ex-post results, we review two recent reports and a causal analysis provided by a NQTT model. As discussed above, the rationale for drawing on these EUKFTA studies derives from: i) the similarity between South Korea and Japan in terms of trade profiles with the EU; ii) the similarity in the text of the EUKFTA and EUJEPA as they belong to the same family of new generation EU FTAs; and iii) the availability of detailed ex-ante and ex-post studies of the EUKFTA that entered into force in July 2011. Such ex-post evaluations are not currently available for CETA[17], EUSFTA[18] or TTIP[19]. We further find reassuring that our expert interviews (see Annex IV for a full list) corroborated our approach, underscoring the high degree of comparability between the EUJEPA and EUKFTA.

Comparison of models

Most studies on the EUJEPA assume a near-complete elimination of tariffs. This seems reasonable given the provisions of the agreement, as discussed in sections III and IV. NTBs are more difficult to pin down. The initial levels and reductions in NTBs can be set based on expert judgement. An alternative method is employed by two NQTT models (Felbermayr *et al*, 2017; Felbermayr *et al*, 2018) that conduct ex-post evaluations of the EUKFTA to approximate trade cost changes for EUJEPA.

Brexit and CPTPP-11 are incorporated in the two Ifo studies. The first Ifo study (Felbermayr *et al*, 2017) models a Brexit that uniformly raises trade costs by 26.5 %[20] between the UK and EU across all sectors. The second Ifo study (Felbermayr *et al*, 2018) updates this assumption. It increases trade costs between the EU and the UK in accordance with a hard Brexit, ie a return to levels of tariffs and NTBs observed with other WTO members.

A common limitation of CGE models is that the impact on FDI flows is not explicitly analysed. Given Japan's investment-led trade strategy, this is a major shortcoming. Lee-Makiyama (2018) emphasises the importance of this investment channel, stating that EU investments in Japan already generate twice the return rate of EU FDI in other markets. Furthermore, the simulation results do not typically reflect dynamic

[17] Entered into force provisionally in September 2017.
[18] Awaiting approval from the European Council as of May 2018.
[19] Last negotiation round held in October 2016.
[20] This is the trade cost change implied by Head and Meyer (2014) that conduct a meta-analysis of trade policy effects using structural gravity models.

welfare gains from increased innovation activity that might arise following closer cooperation between EU and Japan in emerging industries.

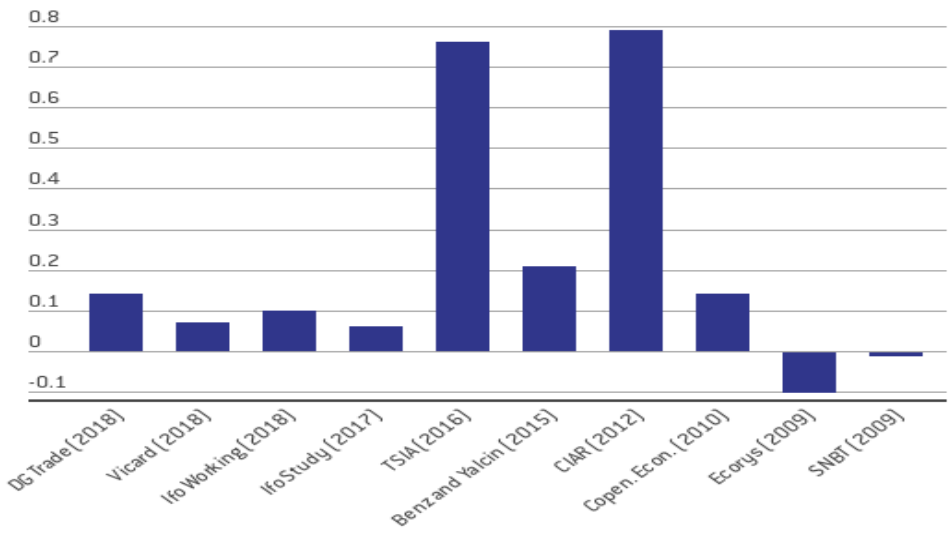
Comparison of economic impact

In what follows, we summarise some of the results originating from the variety of studies on the EUJEPA.

Real GDP Initial studies estimated a minor or negative impact of the EUJEPA on real GDP (SNBT 2009 and Ecorys 2009). Of these, the SNBT (2009) model featured a very conservative liberalisation scenario with only tariff reductions but no NTB elimination between EU and Japan. These estimates have been revised significantly upward with the CIAR (European Commission, 2012) (0.34-0.79 %) and TSIA (European Commission, 2016) (+0.76 %) being the most optimistic. In a recent study, DG Trade (European Commission, 2018) reported an increase of +0.14 % in EU's GDP following the implementation of EUJEPA. If we consider EUKFTA as a good proxy for the trade-cost reductions under EUJEPA, then the effects are relatively smaller in comparison to DG Trade (2018), +0.06 % in Ifo (2017) and +0.10 % in Ifo (2018). New estimates from Ifo (2018) and DG Trade (2018) deviate significantly from the expected GDP growth in the CIAR (2012) and TSIA (2016). This variation in outcomes might be the result of assumptions regarding NTB liberalisation. Over 90 % of the growth in the CIAR (2012) is driven by spill-over effects from reducing two-thirds of all NTBs on an MFN basis. In contrast, DG Trade (2018) assumes that EUJEPA leads to a 3 % reduction in ad-valorem tariff equivalents (AVEs) of MFN service barriers and NTBs are lowered bilaterally in Ifo (2018).

The latest ex-post study on the EUKFTA (Civic Consulting and Ifo Institute, 2017) reports an increase of 0.03 %, or EUR 4.4 billion, in the EU's GDP as a result of the agreement. In percentage change terms, this is significantly lower than the GDP projections of CIAR (2012) and TSIA (2016) and closer to the estimates provided by Ifo (2017). All in all, the majority of studies expect the EUJEPA to have a positive and stronger impact on EU's GDP than the EUKFTA, with more recent studies from 2017 onwards placing the resulting growth for EU in the range of +0.06 to +0.14 %.

Figure 12. Projected impact of EUJEPA on the EU's GDP (% change), across studies



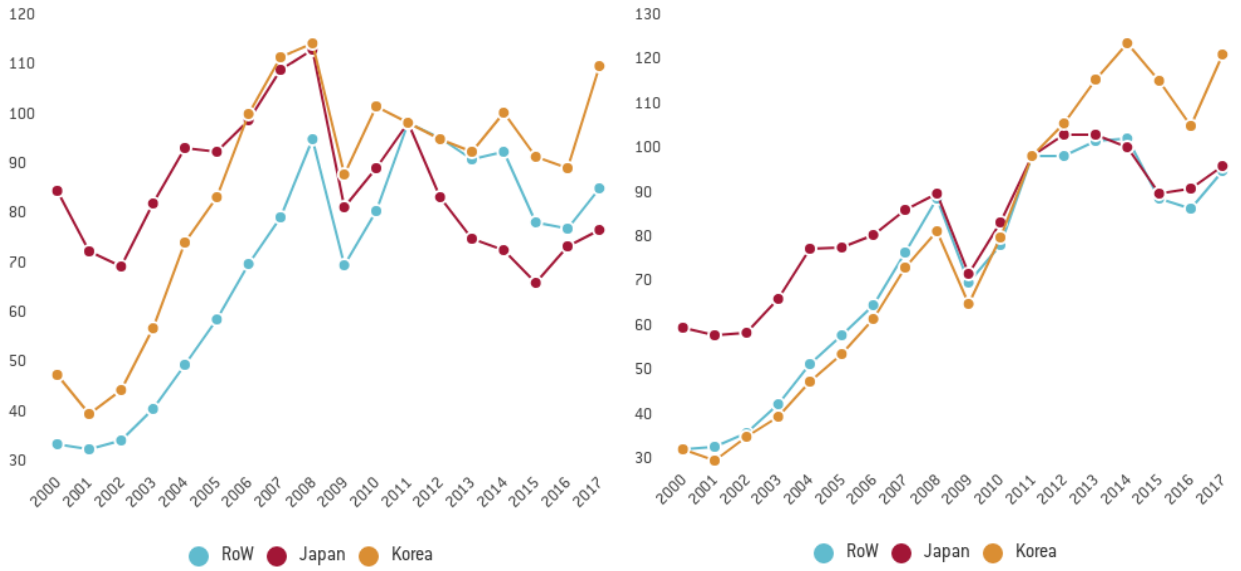
***Source**: Bruegel.*

Notes: Results from the ambitious and asymmetric liberalisation scenario depicted for CIAR (2012) and TSIA (2016). For more details refer to Table 10, Annex II.

Bilateral trade There is substantial variation in the anticipated trade response. DG Trade (2018) found that the EUJEPA would lead to an increase of +13.2 % (EUR 13 billion) increase in EU exports to Japan. EU exports to Japan would rise by 22.6-32.7 % in the CIAR (2012) simulations. A much stronger export response for the

EU is found in Ifo 2017 (61 %) and Ifo 2018 (73 %). This outcome is likely driven by international input-output linkages across sectors. It might also reflect the fact that NTB reductions were modelled as in the case of EUKFTA. On the import side, DG Trade (2018) reports a 23.5 % (EUR 22 billion) increase in EU imports from Japan. This result is similar to that of CIAR (2012) which predicted increases in EU import purchases from Japan in the range of 17.1-23.5 %. In contrast, Ifo (2017) and Ifo (2018) simulations lead to increases in EU imports from Japan of 55 % and 63 %, respectively. We note that the trade response was also strong following the EUKFTA. In terms of observed flows, EU exports to South Korea increased by 59.2 % and imports by 4.8 % from 2010 to 2016. Coming to the causal impact, the EUKFTA led to an increase of 42 % for EU exports to South Korea and 25 % for EU imports from South Korea. Based on these estimates from prior studies, the EUJEPA is expected to lead to a substantial increase in EU-Japan trade that could potentially reverse the declining share of Japan in EU's overall trade, especially in comparison to South Korea since the implementation of EUKFTA in 2011 (see Figure 13).

Figure 13. EU exports and imports to and from Japan, Korea and rest of the world



Source: *UN Comtrade, EU exports and imports to and from Korea and Japan in 2011 normalised to 100.*

Sectoral value added All EUJEPA studies report a positive impact on value added in agri-food industries e.g.+0.82 % according to Ifo 2018, +0.2 % increase in output for processed foods in DG Trade (2018), 0.5-0.6 % for processed foods in the CIAR (2012) and +13 % for meats in Ecorys (2009). For automobiles, projections of value added are model-dependent. Earlier studies such as Copenhagen Economics (Sunesen *et* al, 2009), Ecorys (2009) and SNBT (2009) reported significant contractions of 3 %, 8 % and 6 % respectively for motor vehicles. These effects were revised downwards or turned slightly positive in subsequent studies e.g. -1.6 % in Ifo 2018 and -0.3 % (with tariff reduction) or 0 % (tariff and NTB reduction) in CIAR (2012). The CGE simulations in DG Trade (2018) show a minor contraction in the output of the EU's motor vehicles sector (-0.01 %). By 2035, Japan's car exports to EU increase by nearly 51 % in this study. Because of significant NTB reductions by Japan in motors, the EU also increases its exports to Japan following the EUJEPA (+11.5 %). Hence, the motor vehicles sector is expected to contract under tariff elimination alone but combined with NTB liberalisation, the increase in import competition in EU markets from Japanese car manufacturers would be mitigated by higher EU car exports to Japan. Textiles, apparel and leather products benefit from the agreement as well. The industry is expected to increase its output by 2 % (EUR 7 billion) in the EU, with exports to Japan rising by 220 % (EUR 5 billion) in DG Trade (2018). Looking at the EUKFTA Evaluation study (Civic Consulting and Ifo Institute, 2017), the agreement led to an increase in value added in EU's agricultural sector of +0.3 % and a decline in the automotive sector of

0.19 %. Weighing these projections, we posit that the EUJEPA will significantly increase value added in the EU's agri-food industry and create a minor expansion, or contraction, in value added for the EU's automobile sector[21].

Employment The EUJEPA is expected to have modest implications for employment in the EU. The CIAR 2012 estimated an impact of 0.0001 % to 0.002 % for high-skilled and -0.001 % to +0.002 % for low-skilled workers. The employment losses are more pronounced in the motor vehicles and chemicals industries. The TSIA (2016) adjusts CIAR (2012) projections by +192 000 jobs in manufacturing and services. This is the result of zeroing losses in services, pharmaceutical and chemicals industries that were found to be questionable in its public consultations and sectoral analysis. Benz and Yalcin (2015) also reported an increase of +0.01 % in employment from the EUJEPA under ambitious NTB elimination that prompts within-industry adjustments leading to increased firm sizes and reallocation of labour towards more productive firms. Studies on the EUKFTA shed further light on this issue as ex-ante CGE evaluations of EUKFTA also reported limited employment shifts. On balance, therefore, we expect EUJEPA to produce only modest gains for employment. In the short-run, however, adjustment and reallocation effects could lead to temporary job losses.

Real wages Broadly, positive changes in real wages for low- and high-skilled workers are expected. Moreover, the gains are expected to be similar across the skill spectrum. The greatest gains are reported by the TSIA (2016), according to which real wages will rise by 0.68 % and 0.70 % for low- and high-skilled workers. These estimates are significantly higher than those in the EUKFTA studies. The CEPII[22]/ATLASS (Decreux *et al*, 2010) simulations resulted in wage increases of +0.03-0.04 % for unskilled and +0.04-0.05 % for skilled labour. Similar magnitudes were reported by the EUKFTA evaluation (EU28 average of +0.04 %).

Investment The EUJEPA contains significant commitments on the liberalisation and facilitation of investments, ranging from national and MFN treatment of covered enterprises to anti-trust provisions, intellectual property protection, capital mobility and cross-border movement of mode 4 service providers. Combined with the high levels of investment openness and rate of return in Japan (Lee-Makiyama 2018), the reduction in overall trade costs through EUJEPA is likely to provide a significant boost to bilateral FDI flows. The investment response was significant even under the EUKFTA, despite the absence of explicit commitments on investment promotion. According to the EUKFTA Evaluation 2016, the annual growth rate of the stock of EU FDI in South Korea rose from 5 % in the pre-FTA period to 8 % in the post-FTA years, whereas the growth rate of South Korean FDI in the EU increased from 7 to 19 %. We anticipate a substantial boost to FDI following the implementation of EUJEPA as well. Considering the crucial role of investment in the exchange of technologies, and Japan's leading position in R&D (OECD Innovation Scoreboard), the EUJEPA may further improve EU's growth performance by promoting bilateral investment.

Impact of Brexit Between 2012-2016, the UK was Japan's second largest trading partner within the EU, after Germany, in terms of total value of goods and services traded. Moreover, the Japanese stock of investment in the UK reached EUR 56.7 billion in 2016 with significant investments in the financial sector, transport equipment and professional services. Given this deep trade and investment relationship between the UK and Japan, we can expect Brexit to influence the welfare and trade outcomes from EUJEPA. According to Ifo 2017, the EUJEPA actually becomes more valuable by some EUR 124 million across the EU- 27 Member States following Brexit. There is variation across Member States stemming from differences in the degree of complementarity in comparative advantage with the UK. For instance, the Brexit scenario

[21] Interestingly, several of the experts we interviewed underlined how the risks for European car manufacturers should be considered as relatively limited. This is because: (i) Japan already produces much of its cars inside the EU single market and (ii) many European cars utilise Japanese technological components, the price of which will go down as a result of the EUJEPA.
[22] Centre d'Études Prospectives et d'Informations Internationales (CEPII).

improves welfare outcomes from EUJEPA for Germany as it is expected to substitute for the UK as an export hub within the EU single market. However, EU Member States that export to Japan through the UK could see a decline in welfare gains. Brexit has a greater impact on Japan, as it reduces the economic benefits from EUJEPA by 14 % (EUR 1 billion) and 20 % in Ifo 2017 and Ifo 2018, respectively. This is the result of a smaller market size for Japanese companies following Brexit[23].

Impact of CPTPP-11 Ratification of the CPTPP-11 agreement between Japan and 10 Pacific countries may also affect gains from EUJEPA. This scenario is evaluated in Ifo 2018. Simulation results reveal that the conclusion of the agreement leads to slightly smaller positive gains for the EU and slightly higher real-income gains for Japan (from 0.308 % to 0.314 %) compared to the simple baseline scenario. CPTPP-11 is expected to reduce Japan's costs of sourcing inputs from the Asia-Pacific, improving its competitiveness and trade with the EU.

Distribution of gains It is important to note the pattern of distribution of welfare gains from the EUJEPA. In the analysis by DG Trade (2018), no EU sector is expected to face noticeable losses. In fact, considerable output and export increases are foreseen in industries such as agriculture, beverage, textiles, apparel and leather products that have high rates of SME participation in trade. In TSIA (2016), five sectors – food and processing, other manufacturing, chemicals, business services and motor vehicles – account for 90-93 % of the export gains for both the EU and Japan. This in turn drives the geographic distribution of gains, since countries such as France, Italy and Spain are leading exporters of food and feed to Japan. All EU countries are expected to benefit in simulations conducted by Ifo (2018). Given the expansion of opportunities in agri-food, Italy is expected to benefit more than Germany in their model. Analysing EUKFTA, all Member States benefited, with smaller countries gaining more, according to the EUKFTA Evaluation report (Civic Consulting and Ifo Institute, 2017). In per-capita income terms, Slovakia and Slovenia benefited by EUR 20 and EUR 17 respectively. These gains were similar to Germany (EUR 19) but greater than France (EUR 10) and Italy (EUR 5). Even in terms of overall sectoral value added, EUKFTA had a positive impact on all EU states with Germany, France and Italy as main drivers. Thus, literature-based economic projections indicate that the EUJEPA is expected to have a balanced impact on the economy with all EU countries expected to benefit from the agreement, albeit to varying degrees.

Suggested channels for welfare gains The studies reviewed here suggest several potential channels through which benefits materialise. In CIAR 2012, the overwhelming majority of gains (~90 %) arise from the MFN reduction in NTBs. According to Benz and Yalcin (2015), the benefits stem from higher average firm productivity driven by entry and expansion of more productive firms alongside the exit of less productive firms within industries in both EU and Japan. In Ifo 2018, reduction of NTBs in the services sector is responsible for 73 % of welfare gains in EU. Similarly, in Copenhagen Economics (2009), EUR 22.1 billion or 66 % of the gain to EU's economy is the result of NTB reductions. The contribution of NTB reductions is even steeper for Japan as they account for EUR 15.4 billion or 85 % of the potential welfare gains. All in all, the impact of tariff reductions is relatively smaller given their initially low levels in Japan for industrial goods, as seen in Table 2. Prominent welfare gains through EUJEPA are hence driven by lower NTBs.

5.2 Social impact of EUJEPA

Social concerns such as gender equality, environmental protection, equality in opportunities and inclusive growth are integral to public discourse in trade and have been integrated into the text of trade agreements as well. Trade can also play a vital role in accomplishing targets set in the UN's 2030 Sustainable Development Goals agenda. In this section we review projections on the social impact of the EUJEPA.

[23] While these first order effects remain relevant, dynamically, none of the private sector experts we interviewed saw Brexit as negatively affecting the benefits related to the EUJEPA in a significant manner.

Environment The EUJEPA carries provisions relating to the sustainable management of forests, prevention of illegal timber trade, cooperation in low-carbon technologies and safeguarding marine ecosystems amongst others. Moreover, both parties rank well on the Environmental Performance Index that indicates countries' environmental health and ecosystem vitality. In 2018, Japan is ranked 20th, behind EU Member States such as Austria (8), Finland (10) and Germany (13) but ahead of others such as Greece (22), Portugal (26) and Poland (50).

The implications of EUJEPA for environmental sustainability were addressed in a comprehensive manner by CIAR 2012 and TSIA 2016. In CIAR 2012, the impact on global emissions from the agreement is close to zero (+1.5 million tonnes CO2 or approximately 0.1 to 0.07 % of global baseline emissions). This is largely driven by increased emissions from ASEAN countries due to increased trade alongside a reduction in emissions from China due to trade diversion effects. Moreover, the EUJEPA was expected to marginally lower emissions in Japan by 0.3 % to 0.9 %. To the contrary, TSIA 2016 projected increases in emissions from the EU (+0.28 %) and a decrease in emissions from Japan (-0.14 %). In both parties, the simulations show i) an increase in emissions arising from higher production levels from expanded trade ('scale effect') and ii) a decline in emissions as economic activity shifts from high to low emission producing industries ('compositional effect'). However, the compositional shift is more pronounced for Japan than the EU. Unlike the EU, the compositional effect in Japan is seen to offset the increased emissions from scale effects and thus leads to a contraction in Japan's overall emissions following the implementation of the EUJEPA.

The EUKFTA Evaluation (2017) builds a counterfactual scenario (with no EU Emission Trading Scheme) and reports that additional CO2 emissions from the EUKFTA are low. The study's CGE analysis shows that the EUKFTA led a net global reduction in CO2 emissions by 4.1 million tonnes as the result of trade diversion from China and the US.

Human and Labour Rights Stakeholder consultations conducted for the TSIA (2016) did not reveal any concerns or detrimental consequences for human rights stemming from the EUJEPA. Likewise, the CIAR (2012) did not foresee any direct impact of the EUJEPA on the high standards of protection for human rights in the EU and Japan. Furthermore, the EUJEPA encourages Japan's ratification of two core ILO conventions (Convention 111 on non-discrimination, Convention 105 on forced labour) of which it is not currently part. A detailed assessment of the EUKFTA also revealed that this agreement had a largely neutral impact on the human and fundamental labour rights situation in South Korea (Civic Consulting and Ifo Institute, 2017).

Data Protection In the EU Charter on Fundamental Rights, everyone has the right to the protection of personal data. The collection and processing of this personal information is essential for sectors such as business and financial services, e-commerce and logistics that add further value to other industries in the economy. On 6th September 2018, the European Commission launched its procedure for adopting the adequacy decision on personal data protection with Japan. This involves obtaining an opinion from the European Data Protection Board (EDPB) and consultations with a committee comprising of representatives from all EU Member States. On its side, Japan has extensively modernised the relevant data protection law, the Act on the Protection of Personal Information (APPI), with amendments taking full effect in May 2017. It has also adopted Supplementary Rules that apply to data transferred from the EU. These Supplementary Rules cover the protection of sensitive data, data transfers to third countries and a complaint handling mechanism. EU individuals can complain to an independent Japanese Personal Information Protection Commission (PPC) and file civil action with Japanese courts. Hence, EU individuals will have equivalent rights over their data processed in Japan. Further to this, the European Commission will carry out periodic reviews of the adequacy decision –first, after two years of adoption and every four years subsequently. Within three years of EIF of EUJEPA, the parties will also reassess the inclusion of provisions on free flows of data in the agreement. Given the alignment in data protection regulations between EU and Japan; additional safeguards; avenues for redressal; and institutionalised review and monitoring procedures,

increased cross-border data flows due to trade with Japan should not compromise the right to privacy in the EU.

SMEs The academic literature on trade assumes that a reduction of NTBs lowers fixed costs of trading, leading to an increase on the extensive margin i.e. number of exporting firms. In contrast, eliminating tariffs (a variable cost) leads to adjustment on the intensive margin i.e. volume or value of exports per firm. Therefore, SME participation will depend on the extent to which NTBs are decreased through the EUJEPA. The alignment of Japan with international standards on textile labelling and medical devices along with simplified clearance procedures for SPS measures are examples of this joint effort to reduce NTBs.

As per DG Trade (2018), 88 % of EU firms exporting to Japan are SMEs. Their share in number of enterprises and value of exports is high in agriculture, beverages, textiles and leather but also in construction, real estate activities, wholesale and retail trade. These enterprises are expected to benefit from the agreement, given their substantial presence in EU-Japan trade. The TSIA (2016) draws a similar conclusion, stating that SMEs would benefit from EUJEPA through the deepening of market access especially in agri-food and medical devices sectors. Furthermore, the agreement could help mitigate economic uncertainty faced by SMEs from Brexit and any trade diversion effects from CPTPP-11[24].

Drawing from the EUKFTA experience, we note that the agreement reduced NTBs substantially and lowered fixed costs for SMEs. The EUKFTA Evaluation (2017) provided further statistical evidence of this by using firm-level data on Belgium and Spain available in the World Bank's Exporter Dynamics Database.[25] These data indicate that the EUKFTA benefited Belgian and Spanish firms with low initial sales and enabled new firms to begin exporting to South Korea.

Gender Equality According to the OECD, in 2017, the gender wage gap in Japan and the EU-28 was 24.5 % and 19.1 % of the male wage, respectively. Employment rates amongst women, in percentage of the working age population, stood at 67.5 % in Japan and 62.4 % in EU28. Despite the improvement in women's labour force participation, Japan was ranked 114 out of 144 countries in the Global Gender Gap Index in 2017, far behind EU states such as France (11), Germany (12) and Spain (24).

The EUJEPA is anticipated to have no significant effect on Japan's gender balance through economic channels (TSIA 2016). However, the agreement can lead to an improvement in the policy sphere by encouraging Japan to ratify the ILO Convention on non-discrimination. In CIAR 2012 report, simulations showed that the EUJEPA would lead to a marginal decline (-0.3 %) in women's labour force participation rate in Japan due to increased competition in the services and food processing industries. This can be mitigated by increased investment flows by EU into Japan and a positive evolution of Japan's business climate in favour of gender parity. South Korea's performance in gender outcomes is very similar to Japan's. In 2017, it ranked 118 in the Global Gender Gap Index and had a gender wage gap of 34.7 % of the male wage, the highest amongst OECD economies. The EUKFTA led to little movement in this gender wage gap in South Korea (Civic Consulting and Ifo Institute, 2017). Since 2011, the share of non-regular employment among women has also been fairly stable. Therefore, the EUKFTA had no observable impact in this dimension.

The Japanese government has also introduced a variety of measures to promote women's active participation in the economy as part of its growth strategy under Abenomics. For example, it will increase child care capacity and child care leave benefits to support and encourage the return of women to the labour force after birth. It aims to increase the percentage of women in leadership roles in Japan to 7 % for

[24] Expert interviews suggest that, in order to reap the full benefits associated with the SME chapter, national governments and business associations will have to effectively disseminate information regarding the EUJEPA provisions, which are not necessarily well known to small exporters.

[25] This database provides indicators on the characteristics, sales, entry and exit dynamics of exporting firms of 76 countries over 1997-2014. For further details see https://datacatalog.worldbank.org/dataset/exporter-dynamics-database.

senior government jobs and 10 % for similar positions in companies. The Japanese Ministry of Economy, Trade and Industry has also instituted the 'Nadeshiko Brand' designation along with the Tokyo Stock Exchange (TSE) to publicize TSE-listed enterprises that made important advances in promoting women at the workplace.

On the EU side, gender equality is not explicitly mentioned in the Trade for All strategy. However, new efforts have been launched to mainstream the gender perspective and to understand the gendered impact of trade instruments. In June 2017, the European Commission co-organised the International Forum for Women and Trade along with the International Trade Centre (ITC) to discuss issues such as financing women entrepreneurs and the role of technology in supporting women's participation in global value chains. Most recently in March 2018, the European Parliament endorsed a resolution to account for gender equality in future trade agreements. Subsequent annual reports and impact evaluations of the EUJEPA should also therefore examine the gender dimension in the distribution of opportunities and gains arising from the agreement.

5.3 Opportunities and threats

The various simulation exercises compared under the literature review showed that that the EUJEPA presents multiple export opportunities for EU producers in sectors such as agriculture, processed food, machinery and textiles. At the same time, it exposes EU firms to increased competition in domestic markets from Japanese exporters in motor vehicles, machinery and equipment. As building a full CGE model to estimate the trade impact of the EUJEPA is beyond the scope of this study, we adopt an alternative approach. Using available data, we provide a rough estimate of the distribution of potential gains and losses, or otherwise opportunities and threats, arising from the agreement. Once again, we will do so first in an intuitive simple fashion and later in a more granular and technical manner.

As discussed earlier, the EUJEPA is a comprehensive new generation trade agreement. However, in its simplest form, the largest degree of concessions on the Japanese side is a tariff reduction for what concerns food and wine, together with a recognition of Geographical Indications (GIs), while on the European side is an opening of its automotive market. This allows a rough estimate of the distribution of opportunities and threats from a European standpoint. Regarding the former, Table 6 represents a heat map based on the number of GIs possessed by each EU member state in both agricultural products, and alcoholic beverages. Based on this information, the countries that stand more ready to gain from the opportunities generated by the EUJEPA are Spain, France and Italy, followed by Germany and Portugal. This result confirms the model-based heterogeneity of impact presented in the previous section. Clearly, this analysis shows only the potential direct gains, and does not incorporate indirect effects and positive spillovers, which might well prove beneficial for the other Member States[26].

[26] Another caveat that has been identified during our expert interviews is that GIs are a relatively less known concept in Japan vis-à-vis Europe. As such, realising these gains will require companies to invest in extensive marketing. These benefits are therefore likely to materialise in the longer term and not immediately after EIF.

Table 6. Heatmap based on Geographical Indications for Agri products and alcoholic beverages, by country

	Agricultural products	Wine, spirits and other alcoholic beverages
Austria		
Belgium		
Bulgaria		
Cyprus		
Czech Republic		
Germany		
Denmark		
Estonia		
Greece		
Spain		
Finland		
France		
Croatia		
Hungary		
Ireland		
Italy		
Lithuania		
Luxembourg		
Latvia		
Malta		
Netherlands		
Poland		
Portugal		
Romania		
Sweden		
Slovenia		
Slovakia		
United Kingdom		
Tot EU28 GIs	**72**	**141**

Source: *EU-JP Economic Partnership Agreement.*

Following the same logic as Table 6 above, Table 7 below shows the share of value added in the economy represented by the automotive sector. In a crude way, this information offers a quick snapshot of the countries that would be most exposed to a negative demand shock in this sector, as the one that could materialise following the reduction of tariffs with Japan. From this standpoint, the Czech Republic, Germany, Hungary, and Slovakia are the countries for which the EUJEPA could pose some economic threats. Once again, this depiction should be treated with caution as it represents only direct effects, and treats cars as a homogeneously affected product, while the Japanese car manufacturers tend to have particular characteristics as fuel efficiency, durability and generally smaller-size vehicles rather than design, luxury, and motor performance. As suggested by expert interviews with industry representatives (see footnote 20), and confirmed in the literature review above, most models do not foresee important threats to European car manufacturers stemming from Japanese competition.

Table 7. Total economy value-added, share of manufacture of motor vehicles, trailers, and semi-trailers, by country

	2015
EU28	***1.7***
Austria	1.2
Belgium	0.6
Bulgaria	0.5
Cyprus	0.0
Czech Republic	5.2
Germany	4.5
Denmark	0.1
Estonia	0.6
Greece	0.0
Spain	1.2
Finland	0.3
France	0.5
Croatia	0.1
Hungary	4.9
Ireland	0.1
Italy	0.8
Lithuania	0.3
Luxembourg	
Latvia	0.2
Malta	
Netherlands	0.4
Poland	1.7
Portugal	0.8
Romania	2.0
Sweden	2.3
Slovenia	2.0
Slovakia	4.1
United Kingdom	0.8

Source: *Eurostat.*

We then proceeded to evaluate opportunities and threats in a more granular and technical way. To do so, we collected data on all Most Favoured Nation (MFN) tariffs at the HS 6-digit level for both Japan and the EU. We then computed the Revealed Comparative Advantage (RCA) that both the EU and Japan detain for each product line. This allowed us to precisely identify the areas where Japan holds high tariffs (defined as greater than 10 %) and the EU has a comparative advantage (in line with the literature, defined as greater than 1). We call these 'opportunities', from the European perspective. We also compute the areas where the EU has high tariffs and Japan is a leading exporter worldwide, and call these 'threats', to EU producers.

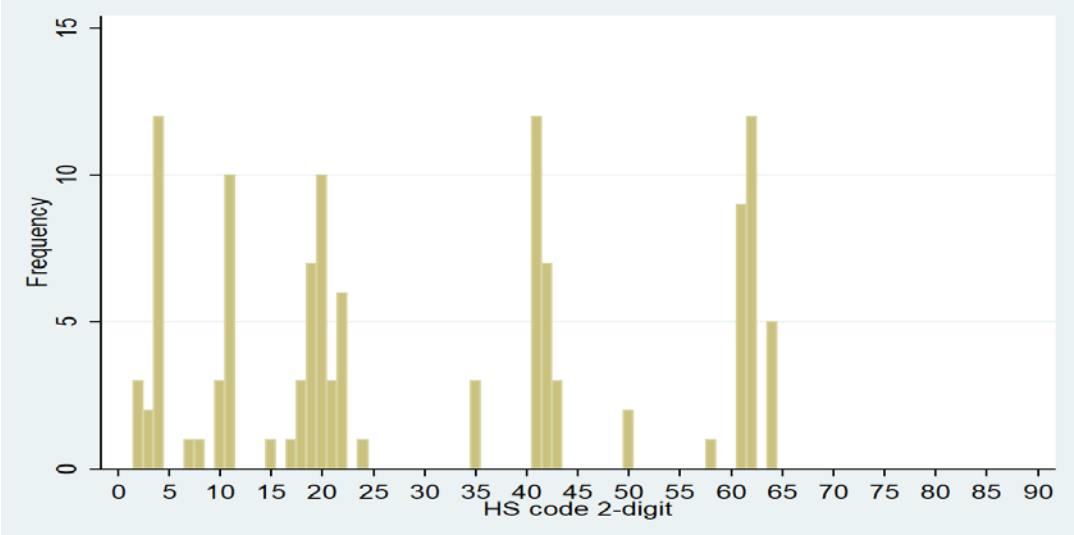
Table 8. HS codes at the 2-digit level

01-05	Animal & Animal Products
06-15	Vegetable Products
16-24	Foodstuffs
25-27	Mineral Products
28-38	Chemicals & Allied Industries
39-40	Plastics / Rubbers
41-43	Raw Hides, Skins, Leather, & Furs
44-49	Wood & Wood Products
50-63	Textiles
64-67	Footwear / Headgear
68-71	Stone / Glass
72-83	Metals
84-85	Machinery / Electrical
86-89	Transportation
90-97	Miscellaneous

Figure 14 below displays the frequency of the opportunities, aggregated at HS 2-digit level to ease readability. Reading this figure in combination with Table 8 above, which contains the legend of HS 2-digit products, leads to interesting results. First, there is a large number of product lines where the EU is a lead exporter and Japan has high tariffs – 118, to be precise. Among these, we notice a cluster between 01-05 and 16-24, which contains inter alia animal products, like cheese, and foodstuffs. This intuitively suggests that the focus of the EUJEPA on these products' tariffs and geographical indications was well placed, and opens ample opportunities for European producers. The other large spikes are at level 41-43, which largely

includes leather goods, and 62-64, which represents clothing and footwear. As tariffs on HS chapters 25-99 are expected to go to zero as a result of the EUJEPA, these areas can be expected to see important opportunities opening up for EU exporters going forward.

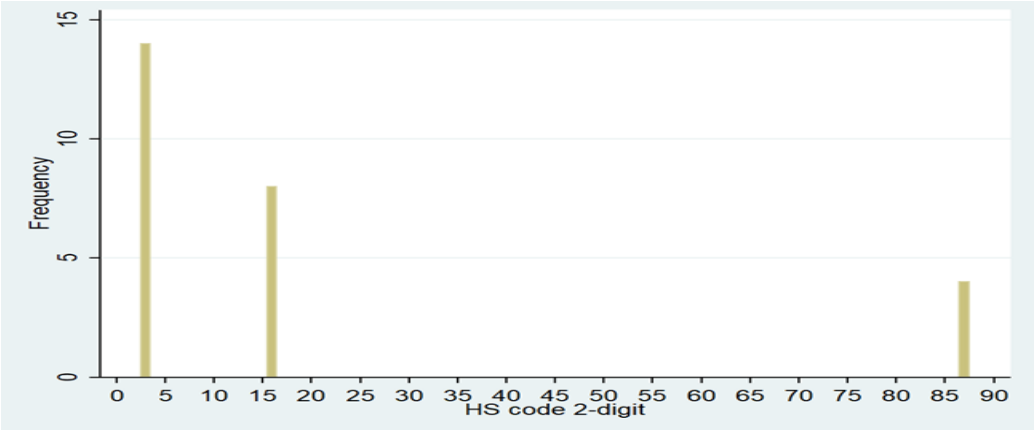
Figure 14. Frequency of opportunities for EU producers based on RCA and MFN tariff levels



Source: *Bruegel calculations based on UNCTAD.*
Note: Opportunities identified as HS 6-digit products where EU RCA>1 and Japan MFN tariffs>10 %.

Along the same logic, Figure 15 below shows the frequency of threats from an EU producers' perspective, aggregated at the HS 2-digit level. The first finding to keep in mind is that the EU has very low MFN tariffs in general. As a matter of fact, there are only 26 product lines (at the extremely granular HS 6-digit level) where the EU had tariffs above 10 % and Japan has a comparative advantage. This explains the relative emptiness of the threats chart against the opportunity chart. The first two peaks (at 03 and 16) identify items that are unlikely to represent any sort of threat to European producers and are typically associated with the Japanese cuisine, such as seaweed, sea urchin, sea cucumber, abalone, and so on. The other peak is around level 87 and, as expected, it relates to motor vehicles.

Figure 15. Frequency of threats for EU producers based on RCA and MFN tariff levels

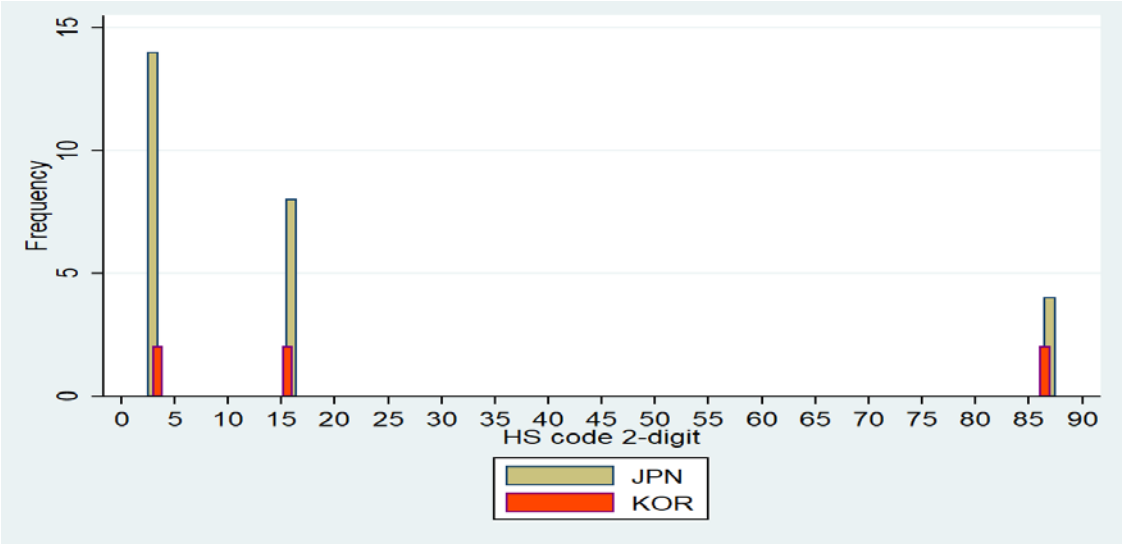


Source: *Bruegel calculations based on UNCTAD.*
Note: Threats identified as HS 6-digit products where Japan's RCA>1 and EU's MFN tariffs>10 %.

As a final check, we reproduced Figure 15 above, identifying however the areas where also South Korea enjoys a competitive advantage and has enjoyed zero tariffs as part of its preferential agreement with the EU. As the EUJEPA is implemented, the introduction of Japanese products in the EU market will likely erode part of this benefit. We will then experience a degree of EU import diversion from South Korea to Japan,

rather than a direct threat to European producers. The red bars in Figure 16 below are therefore to be interpreted as a moderating factor in the analysis of threats.

Figure 16. Frequency of threats for EU producers based on RCA and MFN tariff levels for KOR and JPN



Source: *Bruegel calculations based on UNCTAD.*
Note: Threats identified as HS 6-digit products where Japan's RCA>1 and EU's MFN tariffs>10 %, and HS product categories where also KOR has an RCA>1.

This detailed analysis corroborates our first intuitive findings. Namely, the focus of the EUJEPA on food and wine opens important opportunities for EU agricultural producers. Moreover, the agreement could potentially benefit European leather, clothing, and footwear exporters. At the same time, there are some threats stemming from the opening of the automotive sector to Japanese exporters. All in all, the balance of threats and opportunities from the European side seems to be widely beneficial.

5.4 Non-monetary gains from EUJEPA

The EU and Japan are strategic partners, with shared interests in strengthening democratic rights, the rule of law and an open international economic system. They're important allies for each other and cooperate in numerous fora such as the G7 and G20. The EUJEPA has the potential to deliver strategic benefits that can reinforce this partnership and address common concerns. We discuss the prominent amongst these.

The agreement constitutes an important signal that international trade can stimulate economic growth and produce broad-based welfare gains. It incorporates aspects of social justice, non-discrimination and environmental sustainability in its provisions, serving to address in part a widespread backlash against globalisation.

The agreement unites nearly 640 million citizens of consumers from the EU and Japan under a stable trade and investment regime. It will not only encourage commerce by lowering tariff and non-tariff barriers but can also act as an insurance against economic turbulence by providing regulatory certainty to firms and consumers. Jean, Martin and Sapir (2018) and Vicard (2018) further analyse this insurance dimension, showing that the strategy of establishing trade agreements with multiple partners reduces losses for the EU and its member states by one third in the eventuality of a full-scale trade war[27] and disappearance of the WTO. The reduction in trade policy uncertainty between EU and Japan is an added advantage of the

27 New tariffs on manufactured goods (increase by 60 percentage points) are applied everywhere except within the EU and between its trade partners (Canada, South Korea, Japan, Mexico, Norway, United Kingdom, Switzerland and Turkey). For more details on the trade war scenarios, consult Vicard (2018).

EUJEPA agreement, especially given recent developments such as the US-China tariff dispute and WTO's Appellate Body crisis.

The agreement highlights the advantages of negotiating trade agreements as a bloc to EU Member States. By successfully negotiating a sizeable expansion of market access in Japan, the EU has demonstrated how Member States benefit from their collective negotiating power.

EUJEPA cements the economic partnership and cooperation between the EU and Japan. It represents a significant complement to the SPA that institutionalises joint non-military security efforts spanning disaster relief, cybercrime, energy security and climate change. The agreement can also further cooperation between the EU and Japan in multilateral fora including the World Trade Organisation (WTO) and the United Nations Framework Convention on Climate Change (UNFCCC). At the WTO, closer ties between EU and Japan can enable the resolution of trade issues such as excess capacity in steel, forced technology transfers and non-compliance with WTO's transparency obligations. As the first EU trade deal to carry commitments on UNFCCC, EUJEPA will also promote deeper cooperation between the parties on environmental protection, green growth and renewable energy, thereby developing international standards for sustainable trade practices. It might also help the EU and Japan to contribute to setting digital standards at global level, while at the moment the US (and to some extent China) have taken ownership of this field.

The EUJEPA reinforces a long tradition of EU-Japan cooperation and its various Committees and Working Groups offer new platforms for joint dialogue and policy coordination. As such, these platforms complement endeavours to build political trust alongside diplomatic, cultural and business ties. Other existing agreements between the EU and Japan include the Mutual Recognition Agreement between the European Community and Japan (2002); EU-Japan Agreement on Cooperation on Anti-competitive Activities (2003); Agreement on Mutual Administrative Assistance in Customs Matters (2008); and the EU-Japan Science and Technology Agreement (2009). In addition to these, bilateral cooperation is advanced through the EU-Japan Summit, EU-Japan Business Round Table, EU-Japan Centre for Industrial Cooperation, EU Gateway to Japan programme, EU-Japan Friendship Week, EU-Japan Digital Week, High-Level Japan-EU Development Policy Dialogues and projects financed by EU's Partnership Instrument to improve the understanding of EU in Japan. In the Joint Statement issued at the EU-Japan Summit in July 2018, the partners welcomed new high-level forums of cooperation through the EU-Japan High Level Industrial, Trade and Economic Dialogue[28] and the EU-Japan Policy Dialogue on Education, Culture and Sport.

All in all, our view is that the EUJEPA is a positive and important instrument for the EU and Japan, two leading global players, to reaffirm shared values of transparency, openness and a rules-based economic order against the backdrop of protectionism, isolationism and discontentment with globalisation.

[28] Its meeting will be held on 22 October 2018.

6 Implementation: framework and risks

6.1 Institutional framework

As shown in the overview of provisions (Section III) and through our analysis of legal texts (Section IV), the EUJEPA is a deep and comprehensive trade deal belonging to the family of new generation EU FTAs. Chapter 22 of the EUJEPA lays down the necessary institutional provisions to implement and monitor the progress of implementation of the wide range of commitments under the agreement. These include implementing and tracking tariff and NTB reductions, exchanging information, pursuing regulatory alignment in identified industries and engaging in sustained dialogues on technical standards and intellectual property protection, among other issues.

The agreement explicitly calls for the establishment of:

- **The Joint Committee**: This is a high-level body with a mandate to review and track the implementation of provisions, resolve disputes, set up and supervise or dissolve specialised committees and working groups, make recommendations and provide information to the public.
- **Ten specialised committees**: These are the Committees on Trade in Goods, Rules of Origin and Customs-Related Matters, SPS measures, TBT measures, Trade in Services, Investment Liberalisation and Electronic Commerce, Government Procurement, Intellectual Property, Trade and Sustainable Development, Regulatory Cooperation and Cooperation in the Field of Agriculture. Their distinct functions are defined by the relevant Chapter of the EUJEPA.
- **Two working groups**: These are the Working Group on Wine and the Working Group on Motor Vehicles and Parts, which will operate within the auspices of the Committee on Trade in Goods. Furthermore, ad-hoc working groups may be set up under other Committees, namely SPS, TBT, Regulatory Cooperation and the Joint Committee.
- **Contact points**: These are designated by the parties for the purpose of exchanging information on the contact details of relevant officials and to facilitate communications relating to specific chapters. Contact points have been established for SPS measures, TBT, trade in services, investment liberalisation and e-commerce, government procurement and sustainable development amongst others. Some have a broader mandate, such as the SME Contact Points, which will be responsible for ensuring the needs of SMEs are taken into account in the implementation phase and during policy discussions between parties.

The EUJEPA requires annual meetings to be held by all committees and working groups.

6.2 Implementation roadmap

Trade in goods

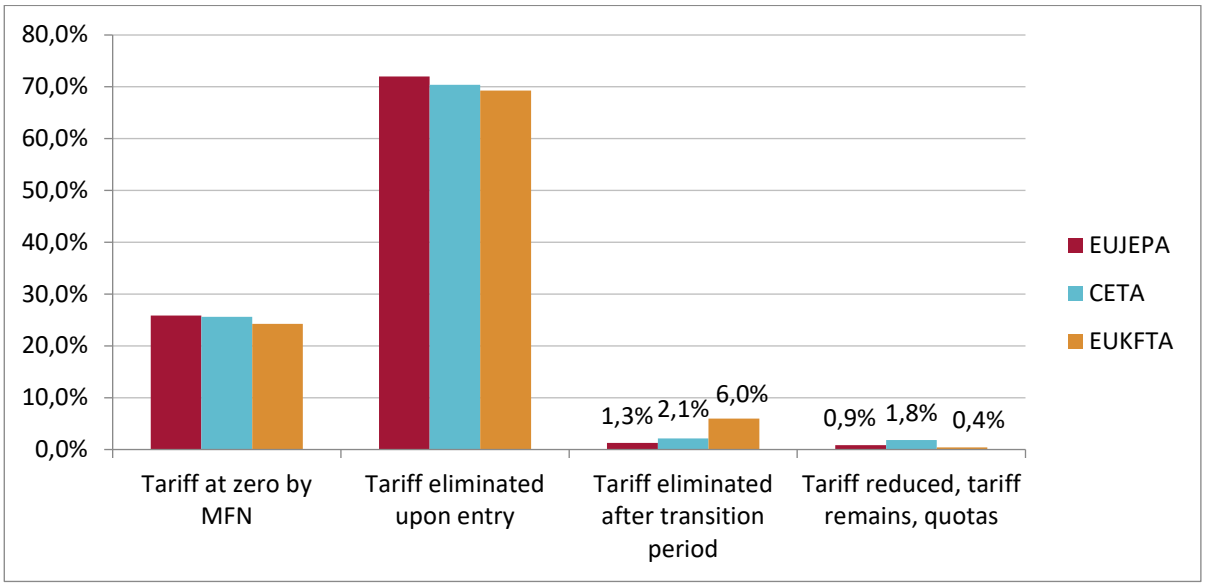
The EU and Japan adopted a negative scheduling approach towards tariffs under this agreement. Therefore, all products not mentioned in the tariff schedules (Annexes 2A) would be duty-free at the entry into force of the agreement. Within the tariff schedules, the EUJEPA contains several different horizons (3, 5, 7, 10, 12, 15 and 20 years) over which duties would be lowered or eliminated.

Figure 17 compares the extent of EU tariff reduction under the EUJEPA to commitments that have been made under CETA and EUKFTA. Even without the EUJEPA, the most-favoured-nation (MFN, Article 1 of the GATT) clause grants companies exporting from Japan to the EU a tariff of zero on 26 % of all tariff lines[29]. Once the EUJEPA enters into force, tariffs on 72 % of tariff lines will be eliminated for Japanese exporters sending goods to the EU. A further 1.3 % of tariffs will be eliminated after a transition period of at most

[29] A tariff line is a product code used for classification of traded goods. We use tariff lines on the 8-digit level of the EU Combined Nomenclature (CN). A tariff line includes several subcategories that can be subject to different tariffs.

15 years. A remaining 0.9 % of tariff lines will remain with a positive tariff, half of which will be reduced from current levels. A comparison across the three agreements shows that the extent of EU tariff liberalisation under the EUJEPA is similar to liberalisation under earlier agreements. However, immediate tariff elimination is granted for a slightly larger share of total tariff lines under the EUJEPA than under the other agreements.

Figure 17. Reduction of EU tariff lines under different free trade agreements



Source: *Bruegel based on data from the WTO Regional Trade Agreements Information System and the Agreement between the European Union and Japan for an Economic Partnership.*

Note: Tariff lines are on the CN 8-digit level. Tariffs are counted as eliminated only if tariffs on all sub-categories are eliminated.

We list some examples of tariff reduction plans below.

On entry into force:

- Japan's tariffs on wine (currently 15 %) and other alcoholic drinks will be eliminated;
- The EU and Japanese tariffs on fishery products to be eliminated;
- Japan's quota system for EU leather and shoe products will be abolished.

Staged reductions:

- Japan's 4.3 % ad valorem duty for high value pork cuts will be phased out over 10 years;
- Japan's tariffs on beef will be cut from 38.5 % to 9 % over 15 years;
- Japan will fully liberalise hard cheeses (current tariffs up to 29.8 %) within 15 years;
- In processed foods, Japan's tariffs will be eliminated over 10 years for pasta, chocolates and confectionery;
- Japan's tariffs on shoes will be cut from 30 % to 21 % over 10 years;
- The EU will fully liberalise tariffs on Japanese automobiles (currently at 10 %) over 7 years.

Trade in services and investment liberalisation

The Committee on Trade in Services, Investment Liberalisation and Electronic Commerce will oversee the agreed elimination of barriers to trade in services and investment. In particular, the Committee will monitor steps taken by national governments to open postal, telecommunication and financial services to enterprises of the partner country. Provisions on the free flow of data are not included in the EUJEPA; however, the need for inclusion of such provisions will be assessed within three years of entry into force of the agreement.

Ten years after entry into force, the treaty foresees a general review of the implementation and operation of the agreement.

6.3 Potential challenges during the implementation phase

We now highlight certain sources of risks/concerns that should be accounted for during the implementation phase of the EUJEPA, considering the long horizons for certain tariff reductions and the scope of the agreement.

1/**Investment liberalisation and promotion** The EU is in separate negotiations with Japan on investment protection standards and the adoption of the EU's reformed mechanism of investment dispute resolution called the Investment Court System, which was included for the first time in CETA. In March 2018, the European Council adopted and published the negotiating directives that authorise the European Commission to negotiate a Multilateral Investment Court (MIC) system that would eventually replace bilateral ICS in existing EU trade and investment agreements. This confirms that the EU position has definitively shifted from the arbitration based ISDS to the court based system in investment dispute resolution. A July 2018 European Commission factsheet on the EUJEPA also stated that for the EU, *'coming back to the old Investor-to-State Dispute Settlement is not acceptable'*[30]. However, ISDS is the mechanism preferred by Japan, one that it has also supported under the CPTPP-11. In the investment negotiations held during July 2018, chief negotiators from the EU and Japan acknowledged the convergence of positions on investment protection standards but not on investor-state dispute settlement[31]. Hence, swift progress on these talks will not be easy but is important to ensure regulatory clarity for promoting bilateral investment under the EUJEPA.

2/**US investigation into auto imports** In March 2018, the US Department of Commerce initiated a National Security Investigation into the import of automobiles and auto parts into the US. This investigation is motivated by Section 232 of US's Trade Expansion Act of 1962 and will examine whether declining American domestic production in the automobile sector poses a threat to its national security by weakening the internal economy and reducing domestic research on advanced technologies. Given the deep value chains in the auto industry, EU and Japanese car producers would be significantly harmed if the investigation leads to an increase in duties on foreign vehicles entering the US market.

On 25 July 2018, European Commission President Jean-Claude Juncker and US President Donald Trump announced plans to hold off on any new unilateral tariffs against each other as bilateral negotiations proceed on liberalising non-auto industrial goods, increasing EU imports of US soybeans and liquified natural gas (LNG), addressing WTO issues, reassessing US steel and aluminium tariffs and EU's imposition of retaliatory tariffs on US goods. The EU will therefore be shielded from the conclusions of the US auto investigations, unless ongoing negotiations are halted by either party. On the other hand, Japan has yet to gain an exemption from a potential increase in US tariffs on autos, with the next round of negotiations planned for September 2018. A coordinated response by the EU and Japan in case of non-exemption could reduce the disruptive effects of a US tariff hike on cars. Hence, the Working Group on Motor Vehicles and Parts might also need to track the evolution of the US investigation and support the exchange of information between parties on this issue.

3/**Auto Safeguards Clause** Annex 2-C of the EUJEPA contains a safeguard clause that permits suspension of concessions and obligations in motor vehicles and parts in case either party shifts away from UNECE standards, reinstates removed NTBs or introduces new NTBs. The safeguard clause is limited to 10 years following entry into force of the agreement, as the EU and Japan deepen their cooperation to develop future standards for this sector. Therefore, the Working Group on Motor Vehicles and Parts should monitor compliance with these commitments in order to flag up risks of a return to MFN tariffs and erosion of preferential access.

[30] http://trade.ec.europa.eu/doclib/docs/2017/july/tradoc_155684.pdf.
[31] EU-Japan investment negotiations, press release: http://trade.ec.europa.eu/doclib/press/index.cfm?id=1888.

4/**Tariff rate quotas (TRQs)** In November 2017, the European Commission released findings from a comprehensive report that assessed the implementation of 25 EU trade agreements of varying depths (European Commission, 2017b). The report showed that TRQs, typically used as market access instruments for sensitive products, are often underused by EU exporters. It stated that for cheese, EU firms utilised only 4.3 %, 7.9 % and 44 % of TRQs for exports to Peru, Colombia and Central America respectively. In the EUJEPA, Japan provides TRQs for agri-foods such as whey products, malt, potato starch, fresh and processed cheeses. In order for these export opportunities to be fully utilised, EU producers, especially SMEs, should receive information and technical support.

5/Japan's future trade ties with the UK According to the draft Withdrawal Agreement between the EU and UK, the UK would be bound by obligations stemming from the EU's international agreements and would be treated as a Member State for the purpose of these agreements during the transition period (March 2019 to end of 2020). The exact terms of EUJEPA would hence be applicable to the UK over this period. Their implementation will be crucial for Japan's car manufacturers that collectively produce 800 000 vehicles in the UK, accounting for 50 % of the UK's total annual production (UK Department of International Trade, 2018). However, after the transition phase, the yet-to-be-negotiated UK-Japan deal would depend on the finalised trade arrangement between the EU and the UK[32]. After 2020, the EU should therefore examine commitments being made in a future UK-Japan FTA in order to prepare for any trade diversion effects under the EUJEPA.

6/ **NTB elimination** According to several of the experts we surveyed, a key challenge in phasing in the elimination of NTBs lies in the fact that the relevant regulations will have to be designed and implemented by ministries (e.g. health ministries in relation to medical devices) that have little contact and experience at international level. Moreover, these ministries were not directly involved in the EUJEPA negotiations. This institutional mismatch might cause some delays. Moreover, NTBs must not only be eliminated, but must also not reappear later down the road. The committee system detailed in section 6.1 was instituted by the EUJEPA to deal with such issues. Experts have however pointed out that this system was largely inspired by the EUKFTA and that, in that setting, committees have proved to some extent to be slow and bureaucratic and have failed to prevent Korea from re-introducing NTBs, such as in the automotive sector.

[32] The current status of Brexit negotiations and the future of UK-Japan economic relations were also discussed in the second meeting of the Japan-UK Trade and Investment Working Group in May 2018.

7 Policy Considerations and Conclusion

After nearly five years of negotiations, the text of the EUJEPA was finalised in December 2017 and submitted by the Commission to the Council in April 2018 as an EU-only agreement, containing only provisions falling under the EU's exclusive competence. The agreement, therefore, only requires EU ratification. On 6 July 2018, the Council adopted a decision on the signature of the agreement and requested the consent of the European Parliament for its conclusion. During the autumn of 2018, the EP's INTA Committee will submit its recommendation to the full Parliament to accept or reject the agreement. Parliament will then decide by a single vote. If the European Parliament gives its consent, the Council will proceed with the conclusion of the agreement.

The two main documents on which MEPs will base their votes for or against the EUJEPA are certainly the December 2017 text of the agreement and most probably the June 2018 assessment prepared by the Commission's DG Trade entitled 'The Economic Impact of the EU-Japan Economic Partnership Agreement (EPA)'.

This report submitted by Bruegel to the European Parliament is an independent evaluation of those two documents and aims to help MEPs form a judgement about the EUJEPA.

According to our analysis, the EUJEPA establishes an ambitious framework to further liberalise and better organise trade relations between two highly developed economies that already have substantial trade and investment ties. This framework, which covers not only trade in goods, but also trade in services, intellectual property and investment, and not only tariffs but also non-tariff measures (NTMs) and regulatory cooperation, belongs to the family of 'new generation' EU trade agreements with developed economies. Other such agreements include the agreements with Canada (CETA), Singapore (EUSFTA) and South Korea (EUKFTA). However, the EUJEPA goes somewhat further than these earlier agreements by introducing new commitments in certain areas such as corporate governance and climate change.

The main findings of DG Trade's assessment of the economic impact of the EUJEPA can be summarised in the following five points:

1. The agreement achieves a high degree of trade liberalisation. The EU will liberalise 99 % of tariff lines and 100 % of imports and Japan 97 % of tariff lines and 99 % of imports. On the 3 % of tariff lines not fully liberalised, Japan has given significant concessions in terms of tariff rate quotas and/or tariff reductions. In addition, Japan will also dismantle a large number of NTMs, including those affecting EU car imports.
2. The agreement has the potential to benefit SMEs thanks to special clauses committing both parties to transparency about market access through the sharing of relevant information.
3. The reduction of tariffs and non-tariff measures foreseen in the agreement (when compared to the situation with no agreement) are expected to add around 0.14 % of additional GDP to the EU by 2035 (when the agreement is expected to be fully implemented).
4. The main export gains for the EU will be in agri-food products and in manufactured goods, including textile, apparel and leather products; for Japan the main gains will be in manufactured products, including industrial goods.
5. No EU sector is expected to experience noticeable losses, including the motor vehicle sector which is expected to see its exports to Japan increase thanks to the reductions in NTMs.

Based on our own qualitative and quantitative analysis, we agree with DG Trade's assessment of the economic impact of the agreement. In a nutshell, the EU and Japan are two high-income countries, with relatively low tariff barriers (except in agri-food products) and highly developed regulatory systems which, though typically aim at protecting consumers, also act at times as NTMs. The removal of trade barriers

(tariffs, NTMs, regulatory cooperation) will therefore benefit both parties and, given their similar income levels, much of the additional trade generated by the agreement can be expected to occur within sectors, with export gains for both EU- and Japan-based producers and therefore few losses in production and employment. As our expert interviews confirm, the car sector, where both the EU and Japan enjoy a strong competitive advantage, is a case in point. In services, the gains are also expected to be fairly balanced between the two countries, with few losses. The one area where the expansion of trade between the EU and Japan is expected to be unbalanced is agri-food, for which the EU enjoys greater competitive advantage than Japan and can be expected therefore to reap greater export gains from the agreement.

One of the pieces of evidence that supports our positive assessment of the economic impact of the EUJEPA is a comparative analysis of the EUKFTA. Korea seems a particularly good choice as a comparator for three reasons. First, as already noted EUJEPA and EUKFTA are fairly similar agreements. Second, Japan and Korea have very similar production patterns, implying that the structure of EU-Japan trade is very similar to EU- Korea trade. Third, the EUKFTA has been operational for several years, having been provisionally applied since 2011 and having fully entered into force in December 2015. Japan and Korea trade experts in both the private sector and academia corroborated this approach. In less than two years, the latest ex- post studies suggest the EUKFTA boosted EU GDP by 0.03 % or EUR 4.4 billion, and led to a 42 % increase in EU exports to South Korea.

It is beyond the scope of this report to verify with the help of a model if the figure of 0.14 % of additional GDP for the EU by 2035 is reasonable or not. However, based on earlier published studies of the EUJEPA (which used not the actual liberalisation decided at the end of the five-year negotiation but hypothetical liberalisation scenarios), it appears plausible, though at the high end of the range of past estimates.

Beyond these quantifiable economic benefits, the EUJEPA offers another economic benefit, which is more difficult to quantify but potentially very important. It comes from the fact that, together with the existing agreement with Korea and the agreements under negotiation with other countries in the region, the EUJEPA will boost the economic presence and political relevance of the EU in the Asia-Pacific area, which is likely to host most of the world's economic growth and activity in the years ahead.

Going beyond its economic benefits, the EUJEPA also has important non-economic implications. The EU and Japan share the common values of liberal democracies and social market economies. Moreover, they share a deep commitment to the rules-based multilateral system not only but also in other areas, including climate change. Closer economic and political cooperation thanks to the EUJEPA will reinforce their ability to shape the course of global developments in a manner that better reflects their shared interests and values. This is particularly valuable at a time when global powers such as China and Russia are promoting an alternative value system, characterised by pragmatic authoritarianism and illiberal democracies, and when the United States, a close ally of both the EU and Japan, appears to be less willing than before to uphold the rules-based multilateral system that they helped create during and after the second world war.

Overall, therefore, we conclude that the EUJEPA is a well-crafted agreement that will help promote the economic and non-economic cooperation and interests of both parties, in their bilateral relations and also in the multilateral context.

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9 Annexes

Annex I - Japan's outward FDI by country/region

Table 9. Japan's outward FDI by country/region (USD Million, %)

	2015	2016	2017	Share	Percent change	Jan-May, 2018 (P)	Share	Percent change
Asia	35,057	13,745	38,266	22.7	178.4	19,830	32.1	59.9
China	10,011	9,453	9,679	5.7	2.4	3,914	6.3	14.4
South Korea	1,593	1,626	1,700	1.0	4.5	1,061	1.7	153.7
ASEAN	20,920	-5,340	22,011	13.1	-	12,214	19.8	88.9
Singapore	7,010	-18,581	9,677	5.7	-	7,326	11.9	305.2
Thailand	4,057	4,632	4,724	2.8	2.0	2,724	4.4	18.7
Indonesia	3,213	2,957	3,388	2.0	14.6	1,136	1.8	5.9
Malaysia	2,918	1,394	935	0.6	-32.9	△ 347	-	-
Philippines	1,531	2,319	1,006	0.6	-56.6	257	0.4	-54.9
Vietnam	1,446	1,672	2,001	1.2	19.7	888	1.4	8.2
India	-1,041	4,105	1,060	0.6	-74.2	1,579	2.6	72.5
North America	51,451	53,327	52,879	31.4	-0.8	302	0.5	-98.6
US	50,218	52,584	51,981	30.8	-1.1	△ 1,330	-	-
Latin America	6,973	27,965	10,950	6.5	-60.8	15,156	24.6	56.7
Mexico	1,229	1,872	1,201	0.7	-35.9	677	1.1	-
Brasil	-193	898	-3,593	-	-	923	1.5	-
Oceania	6,669	6,344	3,185	1.9	-49.8	3,030	4.9	1055.3
Australia	5,676	4,696	2,213	1.3	-52.9	2,568	4.2	-
Europe	36,081	72,157	59,536	35.3	-17.5	22,564	36.6	-38.1
EU	35,785	69,122	56,845	33.7	-17.8	20,328	33.0	-43.5
UK	13,979	49,983	21,628	12.8	-56.7	12,246	19.9	-2.6
World	138,428	173,855	168,587	100.0	-3.0	61,692	100.0	-24.0

Note: 1) The yen-based value is converted to dollars by quarter, using the average quarterly Bank of Japan interbank rate.

2) For after 2014, figures reflect the annual revision. 3) The cumulative total for 2018 is a preliminary figure.
Source: "Balance of Payment Statistics" (Ministry of Finance, Bank of Japan).

Source: *JETRO 2017.*

Annex II - Studies on EUJEPA

Table 10. Comparison of models across studies

Study	Model	Inclusion of DDA conclusion	Inclusion of EU FTAs with South Korea, ASEAN, India, Canada	Tariff elimination	NTB estimation	Liberalisation Scenarios	NTB Spillovers
DG Trade (European Commission, 2018)	Dynamic GTAP CGE	No	EUKFTA, CETA, CPTPP-11	Directly observed tariff rate reductions due to the agreement.	Qualitative assessment of negotiation achievements with scores of 0 (none), 33 (low), 66 (medium) and 100 (high) corresponding to the extent of NTB elimination. Achievement scores then multiplied with actionable NTBs for various HS product categories. 3 % reduction in EU-Japan AVEs of MFN services barriers	Entry into force of EUJEPA	3 % reduction in EU-Japan AVEs of MFN services barriers
Vicard (2018)	Structural Gravity					Gravity model provides estimates of the effect of existing trade agreements. These are used in the counterfactual to analyse the impact of signing the EUJEPA.	
Ifo Working Paper (Felbermayr *et* al,2018)	NQTT	No	EUKFTA, CETA	Complete	Ex-post evaluation of EUKFTA used to calculate implied changes in trade costs.	Scenario I - Complete tariff elimination and NTB reduction modelled to EUKFTA Scenario II - Simulates EUJEPA under a hard Brexit with reversion to tariffs and NTBs observed in WTO with other countries.	No

						Scenario III - Simulates EUJEPA by including a ratified TPP-11 alongside Scenario I.	
Ifo Study (Felbermayr *et* al, 2017)	NQTT	No	Only EUKFTA	Complete	Gravity model based estimation	Scenario I - Tariff only agreement Scenario II - NTB reduction modelled to EUKFTA Scenario III - NTB reduction modelled to the econometric effects for the average FTA.	No
Benz and Yalcin (2015)	Felbermayr et al. (2011, 2012) - search and matching labour market model with Melitz features (monopolistic competition and heterogeneous firms)	No	Only EUKFTA	Complete	Based on average trade creation effect of regional and FTAs. Reduce tariff and NTBs such that trade creation between EU and Japan is 74 % (Felbermayr et al 2013).	Tariff Scenario: MFN tariffs eliminated Comprehensive: Complete elimination of MFN tariffs and lowering of NTBs equivalent to 18.6 % reduction of bilateral ad-valorum trade barrier.	No
CIAR (European Commission, 2012)	CGE	Yes	Yes	Complete	Updated NTB inventory for Japan, but same NTB estimates employed as in Copenhagen Economics, 2009	Asymmetric scenarios - Conservative: 20 % NTB reduction Ambitious: 50 % NTB reduction	Yes, 2/3 of NTB liberalisation on MFN basis.
Copenhagen Economics (Sunesen *et* al, 2009)	CGE	No	No	Complete	Based on business survey and gravity model estimates	Reduction sector specific	No
Ecorys (Plaisier *et* al, 2009)	CGE	No	No	Near complete (except some sensitive agric. tariffs)	Gravity model-based estimation of Ad valorem tariff equivalents without regarding individual NTBs	Reduction of 75 % of service trade barriers and of 2.5 % in NTBs in goods	No
Swedish National Board of Trade (2009)	CGE	No	No	Complete	None	None	No

Table 11. Comparison of simulation results

Study	GDP (in % change)	Bilateral trade (in % change)	Most affected sectors (% change in output or value added)	Employment (in % change)	Real wages (in % change)
DG Trade (2018)	Increase in EU GDP by EUR 34 billion (+0.14 %)	EU exports to Japan increase by 13.2 % EU imports from Japan increase by 23.5 %.	EU: Textile, apparel and leather (+1.9 % or EUR 7 billion); Processed foods (+0.2 % or EUR 1.8 billion); Business services (+0.2 % or approx EUR 13 billion); Motor vehicles (-0.01 % or decline of EUR 140 million). Japan: Motor vehicles (+2.5 % or approx. EUR 13 billion); Machinery and equipment (+1.1 % or EUR 2 billion); Textiles, apparel and leather (-2.2 %) and Dairy (-1.5 %)	n.a.	n.a.
Vicard (2018)	Increase by +0.07 % in EU's GDP per capita	EU goods trade with Japan increases by +0.5 % (in weighted average) for a GDP increase of 0.1 %.	n.a.	n.a.	n.a.
Ifo Working Paper (2018)	Long-run real income increases by USD 15 bn. (+0.10 %) for EU, similar across all scenarios.	EU exports to Japan increase by 73 % (USD 83 bn.) and EU imports from Japan increase by 63 % (USD 79 bn)	Value added in EU: Agri-food (+0.82 % or USD 7 bn); automotive (-1.6 % or decrease of USD 4.6 bn); electronic equipment (+1.07 %) services (increase by USD 13.5 bn)	Ifo Trade Model holds total employment constant, only allows for labour reallocation across sectors.	n.a.
Ifo Study (2017)	Average change for the EU-28 is +0.06 %.	EU exports increase by 61 % and EU imports increase by 55 %	Value added - EU: Pharmaceutical products (+1.5 %); food (+0.25 %) and motor vehicles (0.25 %) Japan: Electronics (+4.5 %) and motor vehicles (+1.5 %); machinery and equipment (2.5 %)	Ifo Trade Model holds total employment constant, only allows for labour reallocation across sectors.	n.a.
TSIA (European Commission, 2016)	Long term impact for EU's GDP is +0.76 %	+34 % for the EU and +29 % for Japan	EU: Food exporters improve sales by 30-40 %; Motor vehicles (<0.1 %)	Adjust CIAR results by +192 000 jobs in manufacturing and services sectors	low skilled: 0.68 % high skilled 0.70 %
Benz and Yalcin (2015)	+0.21 %	EU exports increase by 4.2 % and EU imports	n.a.	Employment rises by 0.01 %	EU wages increase by 0.2 %

		from Japan increase by 11 %.			
CIAR (2012)	Asymmetric scenarios: +0.34 to 0.79 %	EU exports to Japan increase by 22.6-32.7 % EU imports from Japan increase by 17.1-23.5 %.	EU: Electrical machinery (3.5-9.3 %); Processed foods (0.5-0.6 %); Other machinery (0.1-0.6 %) Japan: Other machinery (5.2-11.5 %); Motor vehicles (2.5-3.4 %); chemicals (-3.5 to -11 %); processed foods (-3.3 to -4 %)	Low skilled: -0.001 to +0.002 High skilled: 0.0001 to 0.002 % Electrical machinery for high and low skilled workers (+6.72 and 6.70 %); small increase in agriculture, processed foods, insurance, construction (between +0.1 and +0.2 %)	low skilled: +0.32 to 0.75 % high skilled: +0.31 to 0.74 %
Copenhagen Economics (2010)	+ 0.10 to 0.14 %	EU exports to Japan +46-71 % of EU's 2008 baseline. EU imports from Japan +40- 61 % of Japan's baseline exports to the EU in 2008.	EU: Motor vehicles (-3 %);Transport Services (1 %) Japan: Motor vehicles(+12 %); Other machinery (-6 %)	n.a.	Low skilled: +0.17 to 0.25 % High skilled: +0.17 to 0.25 %
Ecorys (2009)	-0.1 % for EU26	EU26 exports to Japan +0.4 % EU26 imports from Japan +0.4 %	EU: Meats (+13 %); Motor vehicles (-8 %) Japan: Motor Vehicles (+53 %) Meats (-84.5 %)	Low skilled: -7.8 % High skilled: -7.8 %	Low skilled: -0.1 % High skilled: -0.1 %
Swedish National Board of Trade (2009)	-0.01 %	Trade flows increase by 34 %	EU: Pig and poultry meat (3 %), Iron and steel (8 %), Motor vehicles (-6 %) Japan: Motor vehicles (+25 %), Meat (-18 %), Textiles (-9 %)	n.a.	n.a.

Annex III - Studies on EUKFTA

Table 12. Comparison of models across studies

Study	Model	Inclusion of DDA conclusion	Inclusion of EU FTAs with South Korea, ASEAN, India, Canada	Tariff elimination	NTB estimation	Liberalisation Scenarios	NTB Spillovers
EX-ANTE PROJECTIONS							
IBM Belgium (2008)	GTAP CGE model	Doha tariff reduction assumption - a tariff reduction of 40 % on agriculture and the Swiss formula on manufacturing and other primary products.	No	Partial to complete depending on sector and scenario	Expert judgement.	Partial trade: 40 % food tariff reduction, 100 % non-food tariff elimination, 25-50 % reduction in services barriers Full trade agreement: 100 % reduction in food tariffs, non-food tariffs and services barriers	No
CEPII/ATLAS (Decreux *et* al, 2010)	MIRAGE CGE model	Yes, in one baseline scenario.	Yes, in one baseline scenario (all but FTA with ASEAN)	Official schedule	Expert judgement. Sector-by-sector, based on FTA provisions.	60 % cut in NTBs in automobiles, 80 % cut in consumer electronics, 50 % cut in pharmaceuticals, 20 % cut in other industries.	10 % of NTB cut on autos is on MFN basis. Complete 50 % cut on NTBs in pharmaceuticals is on the MFN basis.
EX-POST EVALUATIONS							
EUKFTA Evaluation (Civic Consulting and Ifo Institute, 2017)	NQTT (Ifo Trade Model)	No	No	Directly observed tariff rate reductions due to the agreement.	Estimate total trade creating effects of EUKFTA sector-by-sector. Then knowing the tariff cuts and using trade elasticities, they compute reduction in NTBs.	Implementation of EUKFTA	No

Table 13. Comparison of results

Study	GDP (in % change)	Bilateral trade (in % change)	Most affected sectors (% change in output, value added or trade)	Employment (in % change)	Real wages (in % change)
EX-ANTE PROJECTIONS					
IBM Belgium (2008)	+0.01 % (Full FTA scenario) for EU25	EU exports to South Korea increase by +0.25 % (in value terms from 2008 baseline) EU imports from South Korea increase by +1.68 %	Marginal to negligible increases (<+ 0.25 %) in textiles, motor vehicles and services	Limited employment shifts	Unskilled: -0.02 % Skilled: -0.08 %
CEPII/ATLAS (2010)	+0.07-0.08 %	EU exports to South Korea increase by +62.08 (EUR 33 bn) to 82.58 % (EUR 41 bn). EU imports from South Korea increase by +23.06 (~EUR 23 bn) to 38.39 % (EUR 34.4 bn).	Agri-food (+0.03-0.05 %); textile (-2.06 to -2.22 %); cars & trucks (-0.40 to -1.39 %);	Most effects are small. Notable changes are in textile (-2.10 to -2.25 % in skilled, -2.17 to -2.30 % in unskilled); cars & trucks (-0.42 to -1.42 % in skilled and -0.47 to -1.50 % in unskilled); animal products (+0.56 to +0.93 % in skilled, similar for unskilled)	Unskilled: +0.03-0.04 % Skilled: +0.04-0.05 %
EX-POST EVALUATIONS					
EUKFTA Evaluation (2017)	+0.03 % (EU sees an increase in GDP by about EUR 4.4 billion due to the FTA)	EU exports to South Korea increase by 42 %. EU imports from South Korea increase by 25 %.	Value added - EU: Electronics, Machinery and equipment (0.39 %); Agriculture (0.3 %) ; Automotive sector (-0.19 %)	Holds total employment constant. Labour reallocation in EU generates +40 000 jobs in machinery, +12 000 in agriculture, with largest loss in European business services (-39 000).	EU28 average of 0.04 %. Positive for all EU Member States.
Forizs and Nilsson (2016)	n.a.	EU exports to South Korea increased by +47% and imports by +16 % between 2011-2015.	Machinery & appliances (EU exports +21 %, imports by +5.8 %); transport (EU exports +124 %, imports by +24.4 %); food & beverages (EU exports +50 %)	n.a.	n.a.
Annual Report on EUKFTA (European Commission, 2017a)	n.a.	EU exports to South Korea increased by 59.2 % and imports by 4.8 % between 2010-2016.	Motor vehicles (EU exports increased by 244 % in value from 2010-2016, and imports by 53%); services (EU exports increased by 49 % and imports by 32 %).	n.a.	n.a.

Annex IV - List of interviewed experts

Field	Name	Institution	Position
Public sector	Tsutomu Koizumi	Mission of Japan to the European Union	Former Head of the Economic Department
	Yoshinori Nakata	Bank of Japan	Director-General International Department
Private sector	Bjorn Kongstad	European Business Council (EBC) in Japan	Policy Director
	Ryota Tsunemi	Japan Business Council in Europe	Policy Manager
	Patrice Chazerand	DIGITALEUROPE	Policy Director
	Maurus Unsoeld	BMW Group	Director Government and External Affairs
	Louis Hinzen	Food Drink Europe	Senior Manager Economic Affairs
	Louisa Santos	Business Europe	Director for International Relations
Academia	Prof Motoshige Itoh	National Institute for Research Advancement - University of Tokyo	President and Professor
	Prof dr Werner Pascha	Duisburg University	Professor for East Asian Economic Studies / Japan and Korea
	Prof Sébastien Lechevalier	École des hautes études en sciences sociales	Professor and President of the France-Japan Fondation at EHESS
	Prof Michael Plummer	SAIS Bologna	Eni Professor of International Economics

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